INFINITE / AUDIOVISUAL NODE WORKSTATION

The Node *Field* Guide

A friendly tour of Infinite's entire node palette: what each node does, why you would reach for it, a quick recipe to start with, and a picture of the node itself. No engineering degree required.

Platform	macOS, Apple Silicon and Intel
Coverage	Every node type, 28 chapters
Pictures	Real screenshots of every node

FIVE CABLES, FIVE JOBS

Every cable is colour-coded so you always know what is flowing through it. Colours follow your theme; these are the defaults.

Image · Modulator · Audio · Notes · Palette

BEFORE YOU START

What even is Infinite?

Infinite is a node-based studio for making sound and image at the same time. Instead of separate apps for music, visuals, and 3D, you drop blocks called **nodes** onto one canvas and connect them with cables: a synth feeding a filter, a shape feeding a kaleidoscope, a drumbeat feeding a bouncing particle field.

Because everything lives on the same clock, sound and image can react to each other in real time: a kick drum can punch a shape, a camera's motion can bend a synth's pitch, a color can become a chord. There is no "render and wait". You hear and see the result the instant you plug a cable in.

This guide walks through every node in Infinite's palette, 333 spawnable node types, including the 34-filter image effects shelf and the 20-effect audio shelf. For each you get what it does in plain English, why you would use it, a short recipe, and a screenshot of the node as it looks in the app. The two big shared-table shelves, image filters and audio effects, get a compact one-line entry each since dozens of them are close cousins. Chapter Twenty covers the Field language, where you write the behavior yourself, and the newest eight chapters (111 nodes) cover shapes, alpha tools, 3D solids, the looper and Prediction family, code-drawn Sketch nodes, Field Notes, MIDI file and out, the Tracking nodes, mesh and curve operators, and the v0.6 synths, effects, modulators, compositing and 3D tools. The last two chapters cover the app around the nodes and a gallery picture of every node.

The screenshots were captured by the app itself with `scripts/screenshot_all_nodes.py`, so they always match the build. Run it again after a release to refresh them.

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CHAPTER

One

2D Sources & Visual Generators

Every visual patch starts somewhere. These nodes make the first picture, from a photo on disk to a shape drawn from pure math, that everything downstream gets to distort, color, and remix.

Image Source

SOURCE · IMAGE OUT

Drag a photo, PNG, JPEG, TIFF, or BMP straight from Finder onto the canvas and it becomes a live texture. This is the simplest way to get your own artwork, logo, or photography into a patch.

TRY IT drop in a piece of your own art and run it through a Kaleidoscope for an instant looping visual.

`Image Source` → `Kaleidoscope` → `Bloom` → `Output`

Video

SOURCE · IMAGE OUT

Plays back any movie file as a live texture, with a Loop switch and a speed dial, crank speed negative and it plays backwards. The simplest way to get found footage moving inside a patch.

TRY IT set Speed to a small negative value for a slow, unsettling reverse loop.

`Video` → `Glitch` → `Color Adjustments` → `Output`

Video In

SOURCE · IMAGE OUT

Captures a live feed straight from your Mac's camera, the built-in FaceTime HD camera, an external USB webcam, or a Continuity Camera iPhone, with a device picker, a resolution choice, and a mirror toggle for when a front-facing camera reads reversed.

TRY IT mirror the feed and route it through Feedback for an instant infinity-mirror selfie loop.

`Video In (Mirror)` → `Feedback` → `Output`

Syphon In

UTILITY · IMAGE OUT

Pulls a live video feed straight out of another app publishing over the macOS Syphon protocol, OBS, Resolume, TouchDesigner, with no re-encoding. Great for bringing a camera feed or another VJ tool into Infinite's node graph.

TRY IT pipe your webcam through OBS, then Syphon it in and cut yourself out with Remove Background.

`Syphon In (Camera)` → `Remove Background` → `3D Scene`

Shape

SOURCE · IMAGE OUT

Draws crisp, perfectly smooth vector shapes out of pure math, 20 primitives from simple circles and stars to hearts, gears, and blobs, no pixelation no matter how big you scale them. Fill, stroke, and roundness are all wide open.

TRY IT wire an LFO into Size so a circle breathes in and out on its own, then glow it up with Bloom.

`LFO` → `Shape (Size)` → `Bloom` → `Output`

Noise

SOURCE · IMAGE OUT

Generates organic, ever-shifting texture, soft clouds, ridged terrain, cell-like Voronoi and Worley patterns, or plain white static, from six procedural noise algorithms. It's the visual equivalent of static: raw material for almost everything else.

TRY IT feed Noise into a 3D Plane's height so the terrain rolls and drifts like real landscape.

`Noise` → `Displacement` → `3D Plane` → `Render 3D`

Texture

SOURCE · IMAGE OUT

A pattern factory with ten procedural surfaces on tap, bricks, cell clusters, swirling marble, wood grain, wave bands, checkers, clouds, and rocky Musgrave terrain, all generated live rather than loaded from a file, so they can be animated forever.

TRY IT run a Brick pattern through Color Ramp to turn plain masonry into a stylized sunset gradient.

`Texture (Brick)` → `Color Ramp` → `Material`

Ramp

SOURCE · IMAGE OUT

A clean two-color gradient, linear, radial, angular, diamond, or spiral. Perfect as a soft backdrop, a mask, or a blend driver for another effect.

TRY IT multiply a radial Ramp over live video to vignette the edges of frame automatically.

`Ramp (Radial)` → `Blend (Multiply)` → `Video`

Formula

SOURCE · IMAGE OUT

A shader playground with more than two dozen ready-made looks baked in, plasma fields, Mandelbrot and Julia fractals, Truchet tiles, Voronoi cells, phyllotaxis spirals, each with four knobs you can wire up to anything else in the patch. For when you want something nobody else's patch has.

TRY IT route your track's energy level into one of Formula's knobs so the pattern pulses with the mix.

`Audio Analyze` → `Formula (Param A)` → `Bloom`

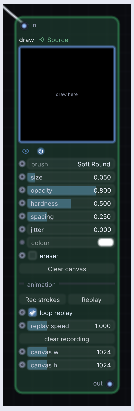
Draw

SOURCE · IMAGE OUT

A live paint canvas right inside the node graph, soft round, hard round, square, spray, chalk, and ink-bleed brushes, plus a dedicated eraser toggle. Anything you paint becomes a texture the rest of the patch can use immediately.

TRY IT paint a quick stroke and route it through Feedback so every mark trails and echoes forever.

Draw Canvas → Feedback → Output



CHAPTER

Two

Typography & Text

Words, on screen or in space. Both nodes read every font already installed on your Mac.

Text

TEXT · IMAGE OUT

Renders crisp type using any font installed on your Mac via CoreText, titles, lyrics, captions. Alignment, size, and letter spacing are all adjustable live.

TRY IT drop a title at the top of your Layer Stack and hit it with Bloom for a glowing lower-third.

`Text` → `Bloom` → `Layer Stack`

Text 3D

3D · GEOMETRY OUT

Takes the same lettering and extrudes it into a real, beveled 3D mesh you can light, rotate, and shade like any other object in the scene, logo stings and title cards with actual depth.

TRY IT give your extruded word a Gold Chrome material and spin it slowly under a studio HDRI.

`Text 3D` → `Material (Gold Chrome)` → `Render 3D`

CHAPTER

Three

2D Image Effects, Blur, Warp & Glow

Take any picture, a photo, a video, a shape, a render, and push it somewhere else. These sit in the middle of a chain, one image cable in and one out, so you can stack as many as you like. The next chapter lists every remaining filter on the shelf.

Gaussian / Box Blur

EFFECTS · IN/OUT

Two separate blur nodes for smoothing an image out, soft focus, depth-of-field, or just taking the edge off something too sharp. Often used right before a Bloom to make the glow itself feel softer.

TRY IT blur at a low radius before Bloom for a dreamy, filmic haze instead of a harsh glow.

`Input` → `Gaussian Blur` → `Output`

Motion Blur

EFFECTS · IN/OUT

Streaks an image in one direction, mimicking a fast camera pan or a subject moving quicker than the shutter can freeze. Instant sense of speed.

TRY IT screen-blend a motion-blurred copy of your video back over the original for a racing-game feel.

`Video Feed` → `Motion Blur` → `Blend (Screen)`

Radial Blur

EFFECTS · IN/OUT

Blurs outward from a single focal point you can place anywhere on screen, the classic zoom-punch effect used to slam attention onto one spot.

TRY IT trigger a quick burst of Radial Blur on every kick drum hit for a punch-in feel.

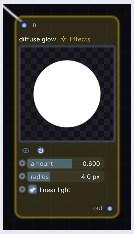
Bloom & Diffuse Glow

EFFECTS · IN/OUT

Two related nodes that make bright areas glow and bleed light into their surroundings, the way a real camera lens does when it looks at something dazzling. One of the fastest ways to make flat visuals feel expensive.

TRY IT drop Bloom right before your Output on almost any visual patch, it rarely looks worse.

`Video / Synth Visuals` → `Bloom` → `Output`



Glitch

EFFECTS · IN/OUT

Six flavors of broken-signal chaos, sliced frames, RGB channel splitting, scanlines, blocky corruption, waving distortion, and full datamosh. Dial it from a subtle VHS wobble to full digital meltdown.

TRY IT fire Glitch's Amount from every drum hit so the visuals corrupt exactly on the beat.

`Audio Analyze (Onset)` → `Glitch (Amount)` → `Output`

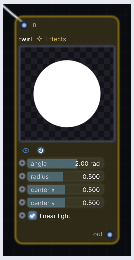
Twirl, Pinch/Punch & Ripple

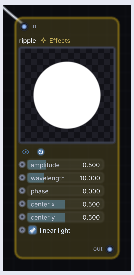
EFFECTS · IN/OUT

Three separate optical-warp nodes: Twirl spins a vortex around a center point, Pinch/Punch magnifies or implodes like a lens, and Ripple sends concentric water rings out from a point. Chain any of them together for compound warps.

TRY IT map your Macro XY trackpad to Twirl's angle and "paint" the swirl live during a performance.

XY Motion Pad → Twirl (Angle) → Live Video





Kaleidoscope

EFFECTS · IN/OUT

Slices the frame into mirrored wedges around a center point, instantly turning ordinary footage into a mandala.

TRY IT point a webcam or Syphon feed straight into Kaleidoscope for an endless, hypnotic VJ loop.

Live Camera → Kaleidoscope → Color Adjustments

CHAPTER

Four

2D Image Effects, The Full Filter Shelf

Infinite's image filters all share one C++ node class, but each of the 34 entries below spawns as its own node in the palette. The seven in Chapter Three got the full treatment; here's the rest of the shelf, one line each, straight from the shader source.

All 34 entries live in one shared table in the app's source; here they are grouped roughly by what they do, distortion & detail, color, and compositing & glow.

Unsharp Mask

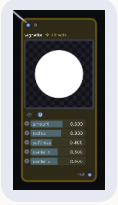
Sharpens the image by comparing it to a blurred version of itself and boosting the difference, controlled by Amount and Radius.

Pixelate

Blocks the image into large flat squares to create a chunky, mosaic look, sized by Block Size.

Add Noise

Sprinkles random per-pixel grain onto the image, with Amount controlling how strong the speckling is.



Vignette

Darkens the corners and edges around a center point, with Amount for strength and Radius/Softness for how far the darkening spreads.

Transform

Moves, scales, and rotates the whole image using Translate, Scale, and Rotation controls, leaving anything pushed outside the frame transparent.

Lens Distortion

Bulges or pinches the image like a camera lens and fringes the edges with color, controlled by Barrel curvature and Chromatic Aberration amount.

Displace

Pushes pixels sideways using either a second connected image as a displacement map or, with nothing plugged in, an animated wavy pattern.

Liquify

Warps the image with a smooth, flowing animated distortion like moving liquid, shaped by Amount, Scale, and Speed.

Symmetry

Mirrors one half of the image onto the other across a chosen axis and center point, with an option to flip the whole result.

Crop

Trims the edges of the image by set Left/Right/Top/Bottom amounts, then either leaves the trimmed area transparent, zooms in to fill the frame, or fills it with a solid color.

Emboss

Turns the image into a raised, carved-relief look by comparing pixels on either side of a light direction, with Amount and Angle controlling depth and lighting.

Convolve

Runs a custom 3×3 grid of numbers over every pixel and its neighbors, letting you dial in blur, sharpen, edge, or emboss-style looks from nine kernel values.

Halftone

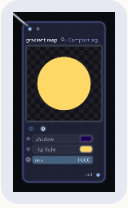
Recreates the image using a grid of dots sized by brightness (like a newspaper print), with Scale controlling dot density.

LUT

Applies a color lookup table by wiring in a second texture as the LUT reference, the same idea colorists use for film-stock emulation.

Lookup

A related color-remap node that reads a reference gradient/table image to redirect each pixel's color.

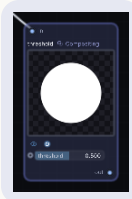


Gradient Map

Recolors the image by mapping its brightness onto a custom two-color (or multi-stop) gradient.

Posterize

Reduces the image to a limited number of tonal steps per channel for a flat, poster-print look.



Threshold

Converts the image to pure black and white based on a brightness cutoff point.

Exposure

Brightens or darkens the whole image in stops, the same control a camera's exposure dial gives you.

Invert

Flips every color to its opposite, turning the image into its photographic negative.

Outer Glow

Adds a soft colored glow that blooms outward from the image's edges or transparency, with Amount for strength and Color for tint.

Color Overlay

Tints the whole image toward a single flat color, with Opacity controlling how strongly the tint replaces the original colors.

Drop Shadow

Casts a soft offset shadow of the image's silhouette behind it, with Offset X/Y for direction and distance and Opacity/Color for its look.

CHAPTER

Five

Color Grading & Tone Mapping

The finishing pass. Once the image is built, these nodes shape its mood, from a quick contrast bump to a full cinema-grade film look.

Color Adjustments

COLOR · IN/OUT

Your all-in-one grading desk: brightness, contrast, levels, color balance, hue, saturation, vibrance, and a black-and-white switch, the same toolkit colorists use, chained together in one node.

TRY IT warm the shadows and cool the highlights for an instant "teal and orange" cinematic grade.

Camera In → Color Adjustments → LUT

Curves

COLOR · IN/OUT

Draggable tone curves for the combined RGB channel plus red, green, and blue individually, the precise, hands-on way to push contrast or introduce a color cast exactly where you want it.

TRY IT pull an S-curve on the RGB channel to punch up contrast on flat, washed-out noise textures.

Grayscale Noise → Curves (S-Curve) → Output

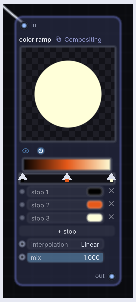
Color Ramp

COLOR · IN/OUT

Recolors any grayscale source by mapping its brightness onto a custom gradient with up to 32 stops, the classic trick for turning plain noise or a height map into something that looks painted.

TRY IT map noise through a five-stop sunset gradient, then use the result as a material's color.

`Noise Texture` → `Color Ramp (Sunset)` → `3D Material`



LUT

COLOR · IN/OUT

Applies a color lookup table to your image by wiring in a second texture as the LUT reference, the same underlying idea colorists use for film-stock emulation, driven from an image input rather than a loaded file.

TRY IT wire a strip-LUT texture into your final Render 3D pass for a consistent, repeatable grade.

`Render 3D` → `LUT` → `Output`

CHAPTER

Six

Compositing & Layering

Where multiple pictures become one, and where you keep a busy graph readable. Everything here speaks the peach Image cable.

Blend

COMPOSITING · IN/OUT

Combines two images with a wide range of blend modes, Multiply, Screen, Overlay, and more, plus an opacity dial. This is how you layer a glowing shape over live video, or double-expose two textures.

TRY IT screen-blend a glowing shape over live video for a lightweight double-exposure look.

`Video (Bottom)` → `Blend (Screen)` → `Output`

Layer Stack

COMPOSITING · IN/OUT

A multi-layer compositor with per-layer blend mode, opacity, and visibility, think of it as a mini layers panel living inside your node graph, for building a full scene from separate pieces without a tangle of individual Blend nodes.

TRY IT stack a 3D background, particles, live video, and a text title into one finished frame.

`Background + Particles + Video + Text` → `Layer Stack`

Fit

COMPOSITING · IN/OUT

Resizes and reflows whatever image comes in to a target width and height, with a mode of Fit (letterbox), Fill (crop to fill), Stretch (ignore aspect ratio), or Native (pass through). A background color and opacity fill any letterboxed edges.

TRY IT feed a portrait-shaped camera or image source into a square canvas using Fill mode so it crops to a clean center square.

`Image Source` → `Fit` → `Output`

Switcher

COMPOSITING · IN/OUT

Cycles automatically among up to 4 connected image inputs on a timed interval (beats or seconds, synced to the transport), with a crossfade control blending the outgoing and incoming input, or a manual toggle to freeze on one chosen slot.

TRY IT wire four different visual "looks" in and set it to switch every 8 beats with a short crossfade for an automatically evolving VJ set.

`4x Image Sources` → `Switcher` → `Output`

Null

COMPOSITING · IN/OUT

A zero-cost reroute point: it simply reports whatever texture is plugged into it as its own output, without rendering anything itself. Meant purely for tidying up cable runs in a busy patch.

TRY IT insert one at a junction where several cables cross so the wiring reads cleanly instead of crisscrossing.

`Image Source` → `Null` → `Output`

Viewport

COMPOSITING · IN/OUT

Functionally identical to Null (a free pass-through, no processing) but drawn with a large canvas, so you can actually inspect the image at a given stage of a patch instead of judging it from a tiny thumbnail.

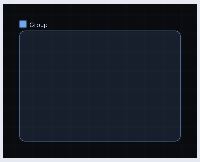
TRY IT drop one mid-chain in a long effects stack to see exactly what the image looks like before the next step touches it.

`Effect Chain` → `Viewport` → `Continue Chain`

Group

COMPOSITING · ORGANIZER

A resizable, colored backdrop for organizing the node editor, not a functional processing node, it has no output of its own. Any node sitting inside it gets dragged along when you move the group's header.



TRY IT wrap a whole "generator" section of a big patch in a Group so you can drag the entire subsystem around at once.

Comment

COMPOSITING · ORGANIZER

A plain annotation box, it does no image processing at all and produces no output. It just holds a block of text plus its own size and background color so you can label or explain parts of a patch.

TRY IT drop one next to a complex chain to leave yourself a note about what a tricky setting does.

CHAPTER

Seven

Feedback, Trails & AI Vision

Nodes that feed an image back into itself, or reach for on-device machine vision, turning one frame into an ever- evolving loop.

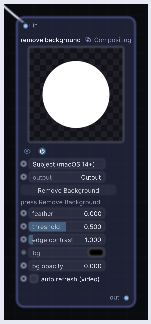
Remove Background

COMPOSITING · IN/OUT

Uses Apple's on-device Vision framework to cut a subject or person cleanly out of a live camera feed, no green screen required. Runs fast enough for live performance, with adjustable feather, threshold, and contrast on the cut edge.

TRY IT drop yourself into a rotating 3D starfield with nothing but a webcam.

`Webcam` → `Remove Background` → `3D Starfield`



Feedback & Trails

COMPOSITING · IN/OUT

Two related nodes that feed a frame back into itself with a slight zoom, spin, or fade, the same trick behind classic video-feedback tunnels. Every new stroke or shape leaves a trail that spirals off into infinity.

TRY IT paint one stroke on the Draw canvas and watch Feedback turn it into an endless spiral.

`Draw / Shape` → `Feedback (Zoom)` → `Bloom`

Reaction Diffusion

COMPOSITING · IN/OUT

Simulates the same Gray-Scott chemistry that grows animal spots, coral, and cellular patterns in nature, feed it a nudge and organic, living patterns spread and evolve on their own.

TRY IT color-map the result and use it as a slowly-crawling texture on a 3D material.

Reaction Diffusion → Color Ramp → 3D Material

Resynthesize

EFFECTS · IN/OUT

An iterative image mutator: instead of reprocessing the original source each frame, it feeds its own previous output back into itself and mutates it again, so the image drifts progressively further from the source over generations. An XY pad cross-fades between eight mutation operators, and its motion can be recorded and looped.

TRY IT record a slow FX-pad path with high feedback and low source-pull, then loop it so the image evolves through the same mutation journey every cycle.

Video / Shape → Resynthesize → Output

CHAPTER

Eight

3D Building Blocks & Materials

Meshes and the surfaces they wear. These nodes speak the green Geometry cable, and everything eventually plugs into Render 3D to become a picture.

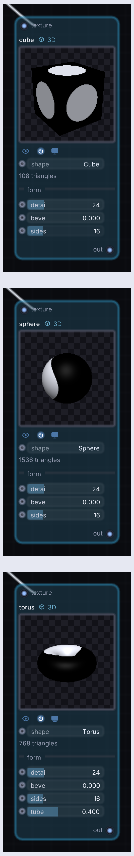
Geometry Primitives

3D · GEOMETRY OUT

A deep library of 3D building blocks, cube, sphere, icosphere, cylinder, cone, torus, plane, disc, and more exotic forms like torus knots and Möbius strips, with adjustable tessellation so they stay smooth up close.

TRY IT drop a sphere, give it a Chrome material, and light it under an HDRI for an instant hero shot.

`Geometry (Sphere)` → `Material (Chrome)` → `Render 3D`



Model 3D

3D · GEOMETRY OUT

Imports your own 3D models, OBJ, PLY, STL, or USD/USDZ, with automatic centering and scale normalization, so anything exported from Blender or elsewhere just drops in and works. (glTF and FBX aren't supported yet.)

TRY IT import a character model and spin it live using a rotation modulated by an LFO.

`Model 3D` → `Material` → `Render 3D`

Material

3D · IN/OUT

Overrides the surface material of whatever geometry passes through it, so one authored look can be shared by several shapes. Exposes a full PBR set, color, metallic, roughness, opacity, emission, IOR, transmission, clearcoat, sheen, iridescence, and can take up to eight image inputs to paint any of those channels from a texture.

TRY IT feed a Noise texture into the roughness map input to give a single flat color a hand-worn, uneven sheen.

Cube → Material → Render 3D

Displacement

3D · IN/OUT

Physically pushes a mesh's vertices using a texture, real 3D geometry displacement, distinct from the 2D Displace/Liquify image filters, which only warp a flat image and never move actual geometry. Offsets vertices by a strength and midlevel, with flat-shading and normal-flip options.

TRY IT drive it with a slow-animated Noise texture to make a flat plane ripple like disturbed water.

Plane → Displacement (texture: Noise) → Render 3D

Mapping

3D · IN/OUT

Changes how a material's texture maps are looked up on a surface without altering the mesh itself, UV, Generated, or Object coordinate space, plus per-axis offset, rotation, scale, and a triplanar-blend control.

TRY IT switch to Object space and animate the rotate values to make a tiled pattern slowly spin across a surface independent of its own transform.

Sphere → Mapping → Material

Wrap

3D · IN/OUT

Conforms one mesh onto the surface of another, every source vertex moves to the nearest point on the target's surface, real shrinkwrap-style deformation. Offers Cylindrical, Spherical, and Nearest Surface modes plus a bend radius and blend amount.

TRY IT wrap a flat text mesh cylindrically around a bottle-shaped target so the text reads as if printed on the label.

`Text 3D + Geometry (target)` → `Wrap` → `Render 3D`

Curve

3D · GEOMETRY OUT

Generates a spline that works two ways: as visible tube geometry (adjustable radius, sides, taper) and as a path other nodes can follow. Builds from up to 8 control points using Catmull-Rom, Bezier, B-spline, or linear interpolation, closeable into a loop.

TRY IT set it to a closed loop and route a Path-follower along it to fly a camera or object in a continuous circuit.

`Curve` → `Path` → `Render 3D`

Null 3D

3D · IN/OUT

A pure pass-through for a geometry cable, forwards the mesh, material, and mapping from its single input unchanged. Exists purely to route or organize a 3D chain, the geometry equivalent of the plain Null node.

TRY IT use it as a labeled junction point where a rig's main geometry chain splits off to several downstream effects.

`Torus` → `Null 3D` → `Material`

CHAPTER

Nine

3D Points, Instancing & Mesh Ops

Scatter, stamp, and clean up geometry at scale, the toolkit for turning one shape into a thousand, or a cloud of points into solid form.

Instance on Points

3D · IN/OUT

Stamps a copy of one mesh onto the points of another, turning a single object into a scattered crowd drawn as GPU instances, one draw call for thousands of copies. Points can come from vertices, edges, faces, or a point cloud, with per-instance scale/rotation randomization and normal alignment.

TRY IT scatter thousands of tiny cubes across a sphere's surface with align-to-normal on so they bristle outward like fur.

`Distribute Points on Faces` → `Instance on Points` → `Render 3D`

Distribute Points on Faces

3D · IN/OUT

Scatters points evenly across the surface area of an input mesh (area-weighted, not vertex-based) so coverage stays uniform even on meshes with uneven face sizes. Offers Random or Poisson Disk (even minimum spacing) methods.

TRY IT scatter Poisson-Disk points across a terrain mesh's faces for evenly spaced grass instances with no clumping.

`Geometry` → `Distribute Points on Faces` → `Instance on Points`

Distribute Points in Grid

3D · GEOMETRY OUT

Generates a plain rectangular lattice of points from scratch, no geometry input needed, with adjustable count, spacing, jitter, and per-point size. Its point ordering deliberately matches Image to Points, so the two line up point-for- point at equal counts.

TRY IT build a 20×20 point grid and instance small spheres on it for a uniform matrix-style array.

`Distribute Points in Grid` → `Instance on Points` → `Render 3D`

Image to Points

3D · IN/OUT

Converts an image into a point cloud, one particle per sampled pixel on a density grid, with brightness pushing each point out of a flat plane by a depth amount, and darker pixels dropped entirely so a cutout image reads as a shape.

TRY IT feed a photo of a face in and instance small spheres on the points to build a pointillist 3D portrait.

`Image` → `Image to Points` → `Instance on Points` → `Render 3D`

Points to Vertices

3D · IN/OUT

Converts an incoming point cloud into a bare mesh made only of vertices, no edges, no faces, so cloud data can re- enter mesh-only operators like Merge by Distance.

TRY IT turn a Particle System's live cloud into vertices, then run it through Merge by Distance to weld nearby particles into clusters.

`Particle System` → `Points to Vertices` → `Merge by Distance`

Set Color

3D · IN/OUT

Writes a color onto every vertex or particle of whatever geometry feeds it, from a flat color, a position/normal-driven gradient ramp, per-element random color, a palette swatch, or a texture lookup.

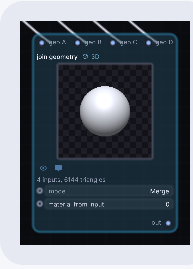
TRY IT feed a Distribute Points node in with source set to Position for a smooth color gradient across a scattered field.

`Distribute Points in Grid` → `Set Color` → `Instance on Points`

Join Geometry

3D · IN/OUT

Merges up to four geometry inputs into a single mesh, using plain concatenation or real CSG Union/Intersect/Difference. Each input's own transform is baked in before combining.



TRY IT feed two overlapping spheres in Difference mode to carve a hole through one with the other.

`Cube + Sphere` → `Join Geometry (Difference)` → `Render 3D`

Merge by Distance

3D · IN/OUT

Welds vertices that lie within a threshold distance of each other, the cleanup pass that closes seams left behind by a scatter or conversion node, or, cranked up, collapses geometry together outright.

TRY IT run Points to Vertices output through Merge by Distance with a larger threshold to fuse nearby particles into solid clumps.

Points to Vertices → Merge by Distance → Render 3D

Metaballs

3D · IN/OUT

Generates blobs that smoothly merge into one another rather than intersecting with hard edges, surfaced via marching cubes over a summed field of up to 8 balls. Can also take a point-cloud input, letting a particle system be surfaced as a liquid-like blob.

TRY IT feed a particle emitter's point cloud in to turn a scattering of points into a single flowing liquid mass.

Particle System → Metaballs → Render 3D

Resynthesize 3D

3D · IN/OUT

The 3D counterpart to the image Resynth node, each generation mutates the previous generation, not the original input, so the mesh drifts progressively over time via displace, jitter, smooth, twist, bulge, extrude, subdivide, and squash operators.

TRY IT set a high displace weight and low smooth weight, then auto-step it to slowly grow an organic, coral-like form from a simple starting mesh.

Icosphere → Resynthesize 3D → Render 3D

CHAPTER

Ten

3D Simulation, Scene & Render

Physics, the camera, the lights, and the render pass that turns geometry into a picture.

Ocean Surface

3D · GEOMETRY OUT

A physically-inspired ocean built from real Gerstner wave mathematics, tune amplitude, wavelength, steepness, direction, choppiness, and speed for anything from glassy calm to a churning storm.

TRY IT drive wave amplitude straight from your track's bass so the sea swells with the low end.

`Audio Analyze (Bass)` → `Ocean (Amplitude)` → `Render 3D`

Cloth Simulation

3D · IN/OUT

Takes any mesh, usually a flat plane, and drapes it like real fabric using position-based dynamics, with stiffness, damping, mass, gravity, and wind you can push around live for anything from a stiff flag to a silky drape.

TRY IT crank wind sideways and watch a plane billow like a flag caught in a gust.

`Plane Mesh` → `Cloth Simulation (Wind)` → `Render 3D`

Particle System

3D · GEOMETRY OUT

Spawns and simulates a stream of moving points with adjustable emit rate, lifetime, initial speed, and turbulence, sparks, dust, snow, or drifting motes, all reacting in real time.

TRY IT trigger a burst of particles on every kick drum for a fireworks-on-the-beat effect.

`Audio Analyze (Kick)` → `Particle System (Emit Rate)` → `Bloom`

Camera

3D · OUT

Defines the viewpoint for Render 3D, perspective or orthographic, field of view, distance/azimuth/elevation orbit around a target, roll, and clip planes. An orbit-per-beat value spins the camera automatically for turntable shots.

TRY IT set orbit-per-beat to a nonzero value for an automatic slow turntable rotation, synced to the transport clock.

`Camera` → `Render 3D`

Light

3D · OUT

Defines a single light source for Render 3D, Directional, Point, Sun, Ambient, or Spot, with azimuth/elevation/distance placement, color, intensity, falloff, and cone angle/softness for spots. Can also orbit automatically over time.

TRY IT use a Spot light with a narrow cone angle and low penumbra for a hard-edged theatrical spotlight look.

`Light` → `Render 3D`

HDRI

3D · OUT

Loads an equirectangular HDR/EXR image and hands it to Render 3D as a real image-based environment, replacing the built-in sky gradient. Exposes intensity and a rotation so you can dial in brightness and spin where reflections and light come from.

TRY IT load an HDRI of a sunset and slowly animate its rotation to sweep studio-quality reflections across a metallic object.

`HDRI` → `Render 3D`

Switcher 3D

3D · IN/OUT

Cycles between up to four connected 3D geometry inputs, forwarding whichever is currently active rather than blending them, a hard cut, timed by beats/seconds or set manually. The geometry-side counterpart to the image Switcher.

TRY IT wire four different Geometry shapes into the four slots and let it hard-cut between them every 2 beats for a stop- motion-style shape change.

`Geometry A/B/C/D` → `Switcher 3D` → `Render 3D`

Audio Displacement

3D · IN/OUT

Deforms a mesh in real time from live audio, with five modes: waveform push, FFT-spectrum warp, concentric ripples, Chladni-plate standing-wave patterns, or a straight axis push. Includes attack/decay smoothing and a real-time subdivision level.

TRY IT patch a kick-drum track in using Acoustic Ripples mode to make a sphere pulse outward on every hit.

`Audio In` → `Audio Displacement (geo: Sphere)` → `Render 3D`

Audio Ribbon

3D · OUT

Builds its own ribbon-shaped mesh directly from an audio signal, no upstream geometry needed. Width, height scale, segment count, gain, and smoothing are all adjustable, alongside its own transform and material.

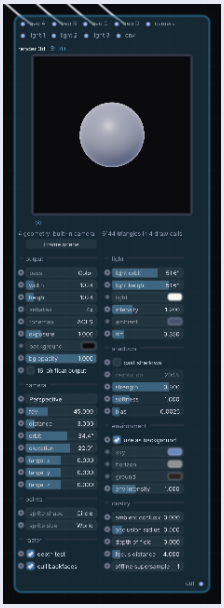
TRY IT route a vocal track in and place several ribbons at different rotations to build a waveform sculpture.

`Audio In` → `Audio Ribbon` → `Render 3D`

Render 3D

3D · IMAGE OUT

The camera and lighting engine that turns every 3D node into a real picture, physically-based Cook-Torrance/GGX materials, ACES tonemapping, and MSAA antialiasing, all live, with up to 4 geometry inputs plus camera, light, and HDRI. This is where the green Geometry cables turn back into a peach Image cable.



TRY IT plug in geometry, a material, a light, and an HDRI, then send the result straight into Bloom.

`Geometry + Material + Light + HDRI` → `Render 3D` → `Bloom`

CHAPTER

Eleven

Audio Synthesizers

Where sound is born. These nodes turn silence, or notes coming in from an arpeggiator or MIDI controller, into audio you can hear and process.

Wavetable Synth

SYNTHS · AUDIO OUT

The flagship synthesizer: 64 wavetables you can morph smoothly between across 8 frames, up to 8-voice unison for thickness, a resonant filter, and three independent ADSR envelopes shaping amplitude, filter, and pitch over time.

TRY IT modulate the table-morph position with an LFO for a synth patch that never quite settles.

Arpeggiator → Wavetable Synth → Reverb

Oscillator

SYNTHS · AUDIO OUT

A single-voice synth on the wavetable engine, offering four waveforms (Sine, Triangle, Saw, Square) with pitch, glide, pitch bend, and an audio-rate FM/PM input whose depth and mode you control directly.

TRY IT patch a second Oscillator into the FM input and sweep FM depth for classic bell/metallic tones.

MIDI Notes → Oscillator → Audio Out

Wave Terrain

SYNTHS · AUDIO OUT

Turns any 2D image or video texture into a "terrain" surface and reads audio by tracing a closed orbit across it, Circle, Lissajous, Lemniscate, Spiral, or Scanline, with full unison, an amp envelope, and a multi-mode filter.

TRY IT feed a slowly-animating Reaction Diffusion texture in and let scan speed drift the orbit for an evolving drone.

`Noise/Shapes Texture` → `Wave Terrain` → `Audio Out`

Equation Synth

SYNTHS · AUDIO OUT

A Desmos-style oscillator: type a formula like y = sin(x + a) and it renders that curve as an anti-aliased, polyphonic wavetable waveform, so turning knobs A through D actually reshapes the wave's math instead of just modulating a fixed one. A full voice sits behind it, unison, an amp envelope, and a resonant filter with its own envelope.

TRY IT start from a plain sine and add a small a * sin(x * 3) term for an instantly richer, custom waveform.

`MIDI Notes` → `Equation Synth` → `Reverb`

Spectral Synth

SYNTHS · AUDIO OUT

Reads an image as a spectrogram: it scans horizontally while each row of pixels defines the amplitude of one of a bank of up to 256 sine partials, so a picture's brightness pattern literally becomes the sound's harmonic content.

TRY IT draw a simple shape in a Draw node and route it in, the silhouette becomes an audible spectral sweep as the playhead crosses it.

Draw/Video Texture → Spectral Synth → Audio Out

Metallic Resonator

SYNTHS · AUDIO OUT

Physically models how real metal and glass actually ring when struck, eight material presets from Steel and Bell Bronze to Glass/Crystal and Vibraphone Bar, a genuinely resonant, inharmonic decay rather than a synthetic approximation.

TRY IT trigger it from Bouncing Balls for a gamelan-like patter of chiming, decaying notes.

Bouncing Balls → Metallic Resonator → Reverb

Sampler

SYNTHS · AUDIO OUT

Load a file or record incoming audio, then play it back one-shot or looping (forward, reversed, or ping-pong) with independent pitch, finetune, and varispeed, and two playback lanes, polyphonic (MIDI-driven) and monophonic (free- running).

TRY IT record a vocal phrase via record-in, then trigger it polyphonically from MIDI Notes with pitch mapped per key.

MIDI Notes → Sampler → Audio Out

Slicer

SYNTHS · AUDIO OUT

Chops a loaded or recorded sample into slices, either by onset detection or by a grid locked to the transport tempo, and maps them chromatically across the keyboard starting at middle C, the classic MPC/Simpler-style chop workflow.

TRY IT record a spoken phrase, slice it by onset, and play the syllables back in a new order from MIDI Notes.

MIDI Notes → Slicer → Audio Out

Granular

SYNTHS · AUDIO OUT

A real-time granular synthesizer that shreds a loaded or recorded sample into micro-grains and re-stitches them, with grain length, density, scan speed/position (with freeze), and stochastic randomization on position, length, pitch, and pan.

TRY IT load a field recording, freeze the scan, and crank the position/pitch randomization for a shimmering granular cloud from a single still moment.

Recorded Sample → Granular → Audio Out

PaulStretch

SYNTHS · AUDIO OUT

Stretches any audio file up to 1000 times longer without changing its pitch, turning a two-second vocal snippet into an hours-long, slowly shifting ambient texture. The classic "everything sounds like a choir of angels" trick.

TRY IT stretch a spoken-word sample 20x and drown it in reverb for an instant soundscape bed.

Vocal Sample → PaulStretch (20x) → Reverb

Molder

SYNTHS · AUDIO OUT

An evolving sound-design tool: it decomposes a recorded sample into tracked harmonic partials plus residual noise, then Roll mutates a parameter genome and re-renders a new sample from that, each roll drifting further from the last, not from the original source.

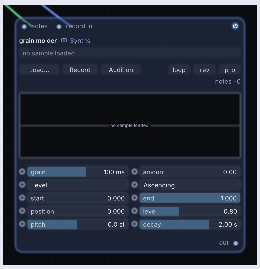
TRY IT record a found sound, then hit Roll a dozen times in a row and listen to it wander into something alien.

Recorded Sample → Molder (Roll) → Reverb

Grain Molder

SYNTHS · AUDIO OUT

Slices audio into overlapping grains, measures each one's loudness, brightness, or a random tag, and re-orders them along a blend between the original chronological order and a fully metric-sorted arrangement, at full amount it turns any recording into a rising or falling contour.



TRY IT sort by Level and crank Amount to turn a busy loop into a smooth, swelling riser.

Audio Loop → Grain Molder (Amount) → Reverb

Drum Sequencer

SYNTHS · AUDIO OUT

An 8-track, 8-step sequencer for your own samples, complete with swing groove, mutes, and solos, build a full beat without leaving the canvas, and tap any lane's trigger to drive visuals too.

TRY IT send the kick lane's trigger into a Glitch node so the picture stutters with the beat.

Drum Sequencer → Audio Out

CHAPTER

Twelve

Audio Effects & Plugin Hosting

Shape sound after it's been made, filters, space, grit, and a doorway to Audio Units already installed on your Mac. Like the image filter shelf, every effect below shares one C++ class but spawns as its own node.

Audio Filter

AUDIOEFFECTS · IN/OUT

A classic analog-style filter with 12 types on tap, low-pass and high-pass at 12/24/36 dB per octave, band-pass, notch, shelves, and peak, with an envelope follower for auto-wah and a sidechain input so another sound can sweep the cutoff for you.

TRY IT sweep the cutoff with an LFO for a slow, breathing filter-sweep synth part.

`Wavetable Synth` → `Audio Filter (LFO Mod)` → `Audio Out`

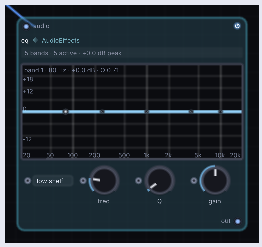
EQ

AUDIOEFFECTS · IN/OUT

A five-band studio equalizer with a live filter-response curve you can see and drag, carve out mud, add air, or fix a harsh frequency, watching the curve move as you turn each knob.

TRY IT scoop the low-mids on your master bus before it hits a Limiter for extra clarity.

`Master Mix` → `EQ` → `Audio Out`



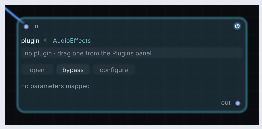
Plugin

AUDIOEFFECTS · IN/OUT

Loads any Audio Unit you already own, synth or effect, right into the graph, with its native interface and up to 32 of its parameters exposed for modulation like any other Infinite node.

TRY IT load your favorite Audio Unit synth and drive it straight from the Arpeggiator.

`Arpeggiator` → `Plugin` → `Audio Out`



Delay

AUDIOEFFECTS · IN/OUT

A tempo-synced ping-pong delay with feedback, tone shaping, panning, and ducking, everything from a tight slapback to a rhythmic, bouncing echo.

TRY IT set the rate division to a dotted-eighth for a classic, rhythmic bouncing echo.

`Melody Synth` → `Delay (1/8 Dotted)` → `Audio Out`

Reverb

AUDIOEFFECTS · IN/OUT

An algorithmic reverb with room size, decay time, damping, pre-delay, and stereo width, a wash big enough to sound like a cathedral, or small enough to feel like a tiled room.

TRY IT stack Reverb after Delay for echoes that themselves bloom into space.

`Delay` → `Reverb (Cathedral)` → `Audio Out`

Drive

AUDIOEFFECTS · IN/OUT

Warm analog-style tube-and-transistor saturation with drive, bias, tone, and color controls, a gentle glow at low settings, real grit when you push it.

TRY IT hit a drum mix with a few dB of Drive for extra punch and glue.

`Drum Mix` → `Drive` → `Audio Out`

Bitcrush

AUDIOEFFECTS · IN/OUT

A separate lo-fi crusher: independent sample-rate (down to 200 Hz) and bit-depth (down to 1 bit) reduction for anything from a subtle vintage-sampler grit to full digital destruction.

TRY IT crush a clean synth line down to 8 bits for an instant retro-game feel.

`Synth` → `Bitcrush` → `Audio Out`

Field Effect

AUDIOEFFECTS · IN/OUT

Runs a Field kernel once per audio sample as a custom effect you write yourself, declare a bit of state for a delay line or filter memory, declare a param for an automatic modulatable knob, and write the DSP math directly instead of hunting for the right built-in. See Chapter Twenty for the Field language itself.

TRY IT write a simple one-pole low-pass in a few lines and compare it against the built-in Audio Filter.

`Audio Source` → `Field Effect` → `Audio Out`

Dynamics

A compressor that squashes loud peaks down once they cross a Threshold, tightening dynamics by a set Ratio, with Attack/Release/Makeup rounding out the feel.

Limiter

A brick-wall ceiling that catches anything above the Threshold and prevents it from getting louder, with In/Out Gain to drive it harder or make up level.

Stereo

Widens or narrows the stereo image with a Width knob (0 collapses to mono), plus Pan and a Bass Mono control that keeps low end centered.

Pitch Shifter

Shifts pitch up or down by semitones (up to two octaves either way) using a delay-line pitch shifter, with Grain size shaping the artifacts.

Chorus

Thickens the sound with multiple slightly detuned, delayed copies, with Delay time and Depth of the wobble and a free or tempo-synced Rate.

Flanger

Creates a sweeping, jet-plane whoosh by mixing a short modulating delay with Feedback back into the signal.

Phaser

Sweeps notches through the sound for a swirling effect, set by a Cutoff center frequency, Depth, and Rate.

Transient Shaper

Reshapes the punch of a sound by independently boosting or cutting the Attack (the initial hit) and the Sustain (the tail).

Stutter

Repeats a short slice of audio like a glitchy loop, with tempo-synced or free timing and a step/gate mask controlling which repeats play.

Ring Mod

Multiplies the signal by an oscillator to produce bell-like, metallic, robotic tones, with adjustable frequency and waveform.

Frequency Shifter

Shifts every frequency in the sound up or down by a fixed Hz amount (not musically, like pitch), creating inharmonic textures, with Feedback adding spiraling "barber pole" motion.

Tremolo

Pulses the volume up and down with an LFO, set by Depth and a free or tempo-synced Rate.

Formant Filter

Gives the sound a vowel-like, talking/singing quality by morphing between A-E-I-O-U formant shapes, with Q controlling resonance.

Wavetable Shaper

Distorts the sound by running it through a chosen wavetable, with Table picking the waveform set, Position scanning through it, and Drive controlling intensity.

The Audio Effects category holds 20 entries sharing one DSP-hosting node class; the seven above got the full treatment, here's the rest of the shelf.

CHAPTER

Thirteen

Audio Utility & Routing

The plumbing that gets sound in, mixed, split, and out, plus two nodes for turning sound into a picture.

Gain

UTILITY · IN/OUT

A simple audio level control: one dB knob raises or lowers the volume of the signal running through it, with an inline meter showing the output level.

TRY IT turn a hot oscillator or sampler down a few dB before it hits a Mixer so it doesn't clip.

`Audio Source` → `Gain` → `Audio Out`

Audio In

UTILITY · OUT

Taps the computer's live input device (microphone or line-in) and brings it into the patch, with a gain trim and its own input-level meter.

TRY IT feed a live microphone into a reactive visual patch so the visuals respond to whatever's happening in the room.

`Audio In` → `Gain` → `Audio Out`

Audio Out

UTILITY · IN

The terminal endpoint of an audio chain, takes exactly one audio input and sends it to the real output device, and can record straight to a file in WAV, FLAC, or MP3.

TRY IT route your final mixed signal here and hit record in FLAC to capture a lossless bounce of a live performance.

`Mixer` → `Audio Out`

Mixer

UTILITY · IN/OUT

The app's dedicated summing node, taking up to 8 input slots, each with its own gain, pan, and mute, plus a per-channel peak meter and an overall summed-output meter.

TRY IT pan several looped instrument stems across the stereo field and mute/unmute them live to build up an arrangement.

`8x Audio Sources` → `Mixer` → `Audio Out`

Splitter

UTILITY · IN/OUT

The explicit fan-out point for audio: since an audio output can only feed one destination, Splitter takes one input and makes it available to up to 4 downstream consumers as a passthrough copy.

TRY IT split one synth's output to feed both a Mixer channel and a separate effects-only chain in parallel.

`Synth` → `Splitter` → `(Mixer + Effect Chain)`

Blend Audio

UTILITY · IN/OUT

A two-input crossfader with a single Blend control: 0 plays only input A, 1 plays only input B, and 0.5 gives an equal- power blend of both.

TRY IT crossfade smoothly between two different loops by automating the Blend knob over a bar.

`Audio A + Audio B` → `Blend Audio` → `Audio Out`

Audio Texture

AUDIO · AUDIO IN, IMAGE OUT

Turns an incoming audio signal into an image, rendering either a live Waveform or a Spectrum (FFT) view into a GPU texture, with adjustable FFT window size and smoothing.

TRY IT feed a bassline in using Spectrum mode with high smoothing for a slowly pulsing frequency-bar texture to layer under other visuals.

`Audio Out (any synth)` → `Audio Texture` → `Compositing Chain`

Audio Color Ramp

COMPOSITING · AUDIO IN, IMAGE OUT

Splits incoming audio into up to 8 frequency bands and maps each one to a color, Intensity mode glows a stop brighter with its band's energy, Expansion widens it, and Spectrum paints a continuous FFT sweep across the palette. Patch an image in and it grades that image through the audio-reactive gradient instead of outputting the gradient alone.

TRY IT leave the image input empty and use it as a standalone audio-reactive gradient background.

`Audio Master` → `Audio Color Ramp` → `Output`

CHAPTER

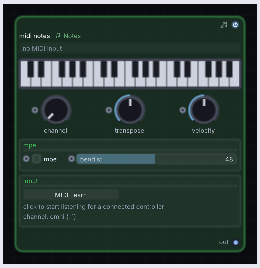
Fourteen

MIDI & Note Shaping

The note stream comes in from a keyboard or controller; these nodes reshape it, filtering, transposing, humanizing, and timing-correcting, before it reaches a synth. All flow over the lavender Note cable.

MIDI Notes

NOTES · OUT



The note source, reads live MIDI input and turns it into note events for the rest of the chain, with a channel selector, a transpose knob, a velocity scale, and an option to snap incoming notes to the global key/scale. An input device dropdown picks which controller it listens to (or all of them), and Learn sets the device and channel from the next key you play.

TRY IT press Learn and play a key on each of two controllers, one Midi Notes per controller, to layer them into different chains without them colliding.

MIDI Notes → Note Transpose → Wavetable

Note Filter

NOTES · IN/OUT

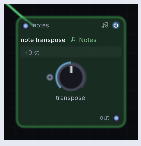
A gate on which notes get through, combining a scale/key filter, a pitch range, and a percent chance, a note only passes if it clears all three checks.

TRY IT set a low chance percentage to turn a busy MIDI part into a sparse, randomly-thinned one.

MIDI Notes → Note Filter → Arpeggiator

Note Transpose

NOTES · IN/OUT



Shifts every note by a fixed number of semitones, with an option to snap the result to the global scale. The shift is applied once at note-on and remembered for that note, so a held note's release always lands on the matching pitch.

TRY IT stack two Note Transpose nodes at +12 and -12 semitones off the same MIDI source to build instant octave doubling.

MIDI Notes → Note Transpose → Sampler

Pitch Bend

NOTES · IN/OUT

A live pitch-bend knob (±2 semitones) that adds into each note's continuous bend amount rather than re-triggering the note, so moving it while a note is held slides that note's pitch in real time.

TRY IT automate the bend knob for a slow pitch-drift or vibrato-like wobble on a sustained note.

MIDI Notes → Pitch Bend → Oscillator

Velocity Curve

NOTES · IN/OUT

Reshapes how hard you hit the keys maps to output velocity, using a single curve exponent, below 1 boosts soft hits, above 1 favors hard hits.

TRY IT set the curve well below 1 to make a light, breathy playing style trigger notes at near-full velocity.

MIDI Notes → Velocity Curve → Sampler

Gate

NOTES · IN/OUT

Controls how long each note rings out with a single hold-time knob. At 0 it's a passthrough; above 0 it ignores the real note-off and schedules its own fixed duration for every note.

TRY IT set a short fixed hold time to turn held chords into a clean, staccato pattern.

MIDI Notes → Gate → Wavetable

Humanizer

NOTES · IN/OUT

Adds random imperfection to a note stream with two independent knobs: timing jitter (when a note actually fires) and velocity jitter (how hard it hits). Both default to off.

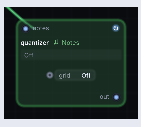
TRY IT add a little timing and velocity jitter to a rigid quantized sequence to make it feel played by hand.

Note Sequencer → Humanizer → Sampler

Quantizer

NOTES · IN/OUT

Snaps each note-on forward in time to the nearest grid line, never earlier than when the note actually arrived, at a chosen musical division.

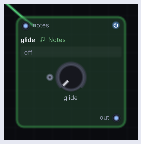


TRY IT use a coarse quantize division to lock loose, off-grid playing onto the beat without losing the original note order.

MIDI Notes → Quantizer → Arpeggiator

Glide

NOTES · IN/OUT



Fills the gap between one played note and the next with a fast run of very short in-between notes, approximating a portamento slide since the note system has no continuous pitch to bend smoothly.

TRY IT set the glide time to around 300–500ms on a bass line for a slippery, portamento-like slide between notes.

MIDI Notes → Glide → Wavetable

Note Echo

NOTES · IN/OUT

Repeats every incoming note a set number of times, each repeat spaced further apart, getting quieter and optionally shifting in pitch. The original note always plays first unchanged.

TRY IT use a short delay with a +12 semitone shift per repeat to turn single notes into an ascending octave echo trail.

MIDI Notes → Note Echo → Wavetable

CHAPTER

Fifteen

Generative & Physical Note Nodes

Nodes that write their own music, from a classic arpeggio pattern to a physics playground that composes melodies from bouncing balls, plus the mergers and switchers that manage several note streams at once.

Arpeggiator

NOTES · IN/OUT

Takes held notes and breaks them into a rhythmic, tempo-locked pattern, Up, Down, Up-Down, Random, As-Played, and a dozen more traversal modes, with 1 to 4 octaves of range.

TRY IT hold a chord from Chorder and let the Arpeggiator turn it into a driving pattern.

Chorder → Arpeggiator (Up-Down) → Wavetable Synth

Chorder

NOTES · IN/OUT

Turns a single held note into a full chord, with adjustable chord size, a strum offset, humanized timing and velocity, and scale quantization so nothing plays a wrong note.

TRY IT play a single key and let Chorder voice it into a lush, gently strummed chord for an instant pad.

MIDI Notes → Chorder → Wavetable Synth

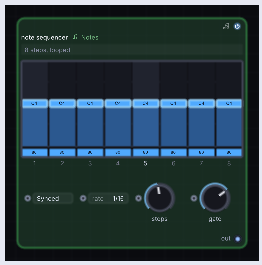
Note Sequencer

NOTES · OUT

A dedicated step sequencer for melody, up to 16 steps, each with its own note, velocity, and on/off toggle, plus a gate- length control for how long each step rings out.

TRY IT program a simple 8-step bassline and send it straight into a synth for an instant groove.

Note Sequencer → Wavetable Synth → Audio Filter



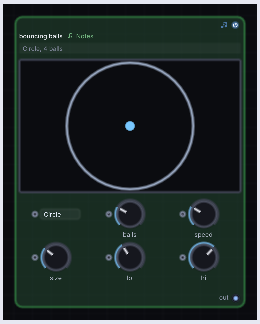
Bouncing Balls

NOTES · OUT

A physics playground that turns bouncing balls into music, balls glide inside a container shape and fire a note, quantized to a musical scale, every time they hit a wall. Gravity doesn't pull them down; the bounce pattern writes the melody.

TRY IT set the scale to Minor Pentatonic and let the balls compose a hypnotic, generative loop.

`Bouncing Balls (Pentatonic)` → `Metallic Resonator` → `Reverb`



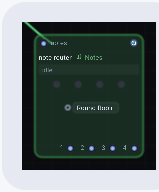
Note Router

NOTES · IN/OUT

The system's note fan-out node: one input split across four separate outputs, chosen by Round Robin, Random, Probability, or Chain mode. A note's matching note-off always replays to the same output it was sent to.

TRY IT use Round Robin mode to alternate incoming notes between four different synths for a call-and-response texture.

`MIDI Notes` → `Note Router` → `(four Wavetable instances)`



Note Merge

NOTES · IN/OUT

The note system's fan-in counterpart to Note Router: merges up to 4 note streams into one output in timestamp order, keeping each input's notes as independent voices so two sources playing the same pitch at once sound as two overlapping notes instead of colliding.

TRY IT merge a live keyboard and an Arpeggiator into one stream so you can play live over your own pattern.

`MIDI Notes + Arpeggiator` → `Note Merge` → `Wavetable Synth`

Note Switcher

NOTES · IN/OUT

The note-cable counterpart of the image and 3D Switcher, cycles between up to four connected note sources on a timed interval or a manual slot, so a synth downstream hears only whichever source is currently active.

TRY IT wire four different sequencers in and let it hard-cut between them every 4 beats for an evolving arrangement.

`4x Note Sources` → `Note Switcher` → `Wavetable Synth`

Random Note Generator

NOTES · OUT

A self-running generator that produces its own note stream via a bounded random walk, each new note is the previous note plus a small random offset, clamped and quantized to a chosen scale and root. No note input needed.

TRY IT set a narrow pitch range with a small max step for a meandering, ambient melodic line that never strays far.

Random Note Generator → Wavetable Synth

Note Stack

NOTES · IN/OUT

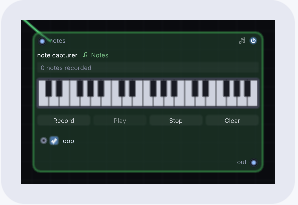
Layers up to 8 extra transposed copies on top of every note that passes through, with the original note always passing through underneath. Each voice has its own semitone offset and on/off switch.

TRY IT enable the +12 and +7 voice offsets to turn single played notes into instant octave-and-fifth power chords.

MIDI Notes → Note Stack → Wavetable Synth

Note Capturer

NOTES · IN/OUT



Records whatever note stream feeds it, with tempo-relative timing so it stays in sync even if the BPM changes, then loops that recording back on top of the live input, which always keeps passing through.

TRY IT record a short improvised phrase, then keep playing live on top while it loops underneath.

MIDI Notes → Note Capturer → Wavetable Synth

Note Strum

NOTES · IN/OUT

Takes a chord's simultaneous notes and spreads their onsets across time in ascending pitch order, like a strummed guitar, while keeping each note's original gate length intact.

TRY IT feed it block chords and set the spacing to 20–40ms for a natural strummed-guitar feel instead of a stabbed chord.

MIDI Notes → Note Strum → Wavetable Synth

CHAPTER

Sixteen

Modulators, Shape & Combine

The wiggling knobs that bring a patch to life. These nodes generate or reshape a signal on the sapphire Modulator/CV cable, which can plug into almost any parameter in the app.

LFO

MODULATORS · OUT

A slow, repeating wave, sine, triangle, saw up, saw down, square, or sample-and-hold, that you plug into almost any knob in the patch to make it move on its own, in time with the music if you like.

TRY IT sync an LFO to a quarter note and wire it to a filter's cutoff for an automatic wobble.

`LFO (Sine, 1/4)` → `Audio Filter (Cutoff)`

Random

MODULATORS · OUT

Generates a new target value on a timer, then eases toward it using a smoothing amount, 0 for hard stepped changes, higher for a fully eased curve. Low/High rescale the range, and each node spawns with its own seed.

TRY IT feed two Random nodes with different rates into separate destinations for uncorrelated wandering motion.

`Random` → `Mod Depth` → `Rotation`

Pattern

MODULATORS · OUT

A 16-step sequencer whose value steps through hand-set slots once per set beat length, looping back to the start, with a smoothing option to ease between adjacent steps instead of jumping.

TRY IT draw a rising-then-falling step shape and route it into a filter cutoff for a rhythmic sweep locked to the beat.

`Pattern` → `Range to Range` → `Filter Cutoff`

Math

MODULATORS · OUT

Combines two other modulator signals using one of ten operations, add, subtract, multiply, divide, min, max, average, and more, then scales and offsets the result.

TRY IT multiply an LFO by a Random node so the wobble's depth itself wanders over time.

LFO + Random → Math (Multiply) → Mod Depth

Compare

MODULATORS · OUT

Tests two modulator inputs against each other using >, >=, <, <=, ==, or != and outputs a hard 1 or 0, with a tolerance value for the equality checks.

TRY IT compare a Random node against a fixed threshold to fire an on/off switch at unpredictable moments.

Random → Compare (A > B) → Mod Depth

Range to Range

MODULATORS · OUT

Linearly remaps an input modulator from one span to a different output span, with an option to clamp the result inside the output bounds.

TRY IT remap an LFO's 0..1 swing down to a narrow 0.3..0.7 band for a subtler wobble.

LFO → Range to Range → Mod Depth

Smoothing

MODULATORS · OUT

Runs an exponential moving average over an input modulator, with an amount knob setting the filter strength, taking the edge off a stepped or noisy source.

TRY IT smooth a Pattern node's stepped output to turn a staircase into a gentle glide.

Pattern → Smoothing → Mod Depth

Invert

MODULATORS · OUT

Mirrors an input modulator around a low/high pivot, so rises become falls and vice versa, and it still behaves correctly on signals that run outside the usual 0..1 range.

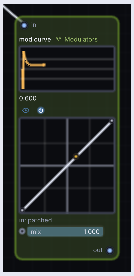
TRY IT invert an LFO feeding one destination while the un-inverted LFO feeds another, so two parameters move oppositely together.

LFO → Invert → Mod Depth

Mod Curve

MODULATORS · OUT

Reshapes an input modulator through a user-drawn transfer curve, with a mix knob blending between the raw input and the fully-curved result.



TRY IT draw a steep S-curve to make a smooth LFO snap more decisively between its extremes.

LFO → Mod Curve → Mod Depth

CHAPTER

Seventeen

Modulators, Control & Performance

Knobs, pads, and hardware bridges meant to be touched live, plus the utility nodes that hold a modulation chain together.

Mod Depth

MODULATORS · OUT

Scales how much of an incoming modulator's swing reaches its destination with a single depth knob from -1 to 1, at 0 the signal has no effect, negative values invert it while still scaling.

TRY IT automate depth itself with a slow LFO to fade a modulation effect in and out over time.

LFO → Mod Depth → Filter Cutoff

Macro Knob

MACROS · OUT

A single manually-set control meant to be performed rather than left alone, with a curve exponent to ease the response in or out and an invert switch.

TRY IT map Macro Knob to several destinations at once as a single "intensity" performance control for a whole patch.

Macro Knob → Mod Depth → Reverb Mix

Macro Slider

MACROS · OUT

A plain 0-to-1 fader, the simplest possible performance control, meant to be grabbed and pushed rather than automated.

TRY IT map it to Reverb Mix for a single fader that wets or dries the whole patch.

Macro Slider → Mod Depth → Reverb Mix

Macro Bipolar Knob

MACROS · OUT

A center-detent knob that swings -1 to +1 rather than 0 to 1, so its resting position genuinely means "no effect", ideal for anything you want to push both directions from a neutral center, like a pan or a pitch bend.

TRY IT center it at rest and map it to a filter's LFO depth so turning either way adds wobble, and dead-center is silence.

Macro Bipolar Knob → Mod Depth → Filter Cutoff

Macro XY

MACROS · OUT

A 2D performance pad, grab it with a mouse or trackpad to fling two parameters around at once, live. It can also record a path you drag out and loop it back for hands-free, repeating motion.

TRY IT map X to a filter cutoff and Y to Glitch amount, then perform both with one finger.

`Macro XY` → `Filter Cutoff (X) & Glitch Amount (Y)`

Macro Toggle

MACROS · OUT

A latching on/off switch, plain and simple, click once to flip it on, click again to flip it off.

TRY IT gate a whole effects chain's Mix on and off with one click during a performance.

`Macro Toggle` → `Blend Audio (Blend)`

Macro Trigger

MACROS · OUT

A momentary bang pad: it fires to 1 the instant you press it and drops back to 0 the instant you release, with its own flash so you can see exactly when it fired.

TRY IT wire it into a Particle System's emit rate for a hand-triggered burst on demand.

`Macro Trigger` → `Particle System (Emit Rate)`

Macro NumBox

MACROS · OUT

A draggable, typeable number box with its own min/max/step range, for a performance control where you want to see and dial in an exact value rather than eyeball a knob's position.

TRY IT set its range to match an LFO's rate in Hz for precise, typed-in tempo control.

`Macro NumBox` → `LFO (Rate)`

Macro Radio Selector

MACROS · OUT

A discrete multi-choice selector, pick one of a set number of options and it outputs that choice's position, evenly spaced across 0 to 1.

TRY IT wire four choices into a Switcher's manual slot for one-click scene selection.

`Macro Radio Selector` → `Switcher (Manual Slot)`

Macro Step Gate

MACROS · OUT

An 8-step rhythmic gate sequencer locked to the transport, toggle any of the 8 steps on or off and it outputs a 1-or-0 pulse in time with the beat, by default at sixteenth notes.

TRY IT draw a syncopated pattern and route it into Glitch's Amount for a rhythmic stutter locked to the beat.

`Macro Step Gate` → `Glitch (Amount)`

MIDI CC

MODULATORS · OUT

MIDI Learn for a single physical control on your USB MIDI hardware, click Learn, twist the knob, and it remembers that device's exact channel and controller number.

TRY IT map a hardware filter-cutoff knob to sweep a Blur radius live during a performance.

`MIDI CC` → `Mod Depth` → `Filter Cutoff`

MIDI Trigger

MODULATORS · OUT

Turns a MIDI pad hit into a decaying envelope pulse (Pad mode, bound to one note), or reports whichever note was last played on the channel (Keyboard mode).

TRY IT trigger a camera-shake or flash burst every time you hit a specific pad.

`MIDI Trigger (Pad)` → `Shape Opacity Flash`

Path

MODULATORS · OUT

Moves a point along a shape over time, Circle, Line, Figure Eight, Helix, Spiral, or Lissajous, outputting separate X, Y, Z, and progress channels, tempo-synced by default.

TRY IT send the X and Y outputs to a Geometry node's position to orbit an object in a circle.

`Path (X/Y/Z)` → `Mod Depth` → `Geometry Position`

Constant

MODULATORS · OUT

Outputs one fixed, unmoving value, exists because math-utility nodes only take other modulators as input, so this supplies a literal number.

TRY IT use it as a fixed offset added to an LFO to keep a modulation from ever reaching zero.

`Constant` → `Math (Multiply)` → `Mod Depth`

Null Modulator

MODULATORS · IN/OUT

A zero-cost pass-through junction for a modulator cable, useful as a named point in a chain where you can read a live value meter or branch one signal to several destinations.

TRY IT insert one mid-chain to visually confirm what value an LFO chain is actually producing.

`LFO` → `Null Modulator` → `(branches)`

Palette

MODULATORS · OUT

Extracts a small color palette from a reference image or live video, clustering colors in Oklab space with sort, chroma, and mood-shaping controls (hue shift, saturation, brightness).

TRY IT feed in a still-life photo and pipe its palette into every Ramp and material color in the patch for one cohesive mood.

`Palette (image)` → `Ramp / Material Color`

CHAPTER

Eighteen

Modulators, Audio/Note Bridges & Analysis

The converters that translate between worlds, audio into control voltage, notes into control voltage, geometry into control voltage, and raw sound or image into a stream of numbers a patch can react to.

CV to Pitch

MODULATORS · OUT

Quantizes a continuous modulation signal into semitone steps, optionally snapped onto a chosen scale and root, with a glide option for portamento smoothing on the stepped output.

TRY IT feed a slow Random node through it set to a major scale for generative, in-key melodic-feeling movement.

Random → CV to Pitch → Oscillator Pitch

Note to CV

MODULATORS · OUT

Converts an incoming note stream's pitch into a normalized control voltage, holding the last played note's pitch rather than falling back to zero on release, a pitch tracker, not a gate.

TRY IT track incoming melody pitch and use it to modulate a filter cutoff so higher notes open the filter.

MIDI Notes → Note to CV → Filter Cutoff Mod

Audio to CV

MODULATORS · OUT

An envelope/amplitude follower that converts a live audio signal into a modulation signal, with gain, attack/release smoothing, and a peak/RMS detection mode.

TRY IT feed a vocal or instrument track in and use the follower's output to pulse a Shape's scale in time with the performance.

Audio File → Audio to CV → Mod Depth

Geometry Table

MODULATORS · OUT

Samples up to 16 points off a patched mesh and reports each one as its own X/Y/Z modulator, plus an overall centroid and spread, turning the literal shape of a piece of geometry into a bank of control voltages. Vertex, Scatter, and Contour sampling modes read different things off the source, and an offset control scrubs a read head across it without re- sampling.

TRY IT sample 8 points off a spinning Torus and drive 8 separate effect params straight from its shape.

`Torus` → `Geometry Table` → `(8x Mod Depth)`

Vibrato

MODULATORS · OUT

A simple free-running sine LFO with one control, rate in Hz, meant as a substitute for true per-note vibrato, since note events have no continuous pitch channel of their own.

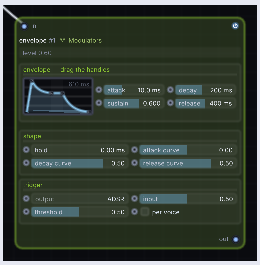
TRY IT patch it into an oscillator's fine-tune modulation input for a subtle, continuous pitch wobble.

`Vibrato` → `Oscillator Fine-Tune`

Envelope

MODULATORS · OUT

Applies a classic ADSR contour to a continuous modulation signal rather than a note trigger, the "gate" is derived from the input crossing a threshold rather than a note-on/off.



TRY IT feed a slow LFO in to get a rhythmic swell-and-release pulse instead of a raw sine wave.

`LFO` → `Envelope` → `Mod Depth`

Audio File

MODULATORS · OUT

Loads and plays back an audio file independently of the live mic/line input, exposing the same analysis taps as live input so a backing track can drive visuals deterministically.

TRY IT load a pre-mixed track and drive the whole visual patch from it so a live performance is reproducible.

`Audio File` → `Audio Analyze` → `Mod Depth`

Audio Analyze

MODULATORS · OUT

Listens to any audio and splits it into 8 frequency bands plus a transient onset trigger, the node that lets visuals genuinely react to music, band by band, hit by hit.

TRY IT route a bass band into the Ocean node's amplitude so the sea pulses with the low end.

`Audio Master` → `Audio Analyze (Bass)` → `Ocean (Amplitude)`

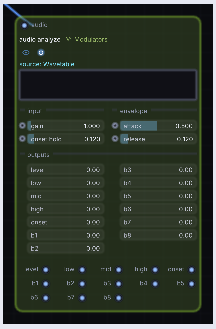


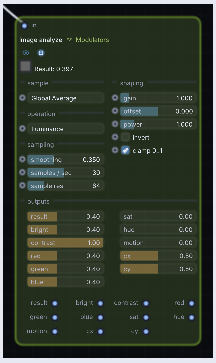
Image Analyze

MODULATORS · OUT

The visual mirror of Audio Analyze, reads brightness, color channels, saturation, hue, motion, and where the action is happening on screen, turning all of it into modulation.

TRY IT drive a synth's pitch or filter straight from webcam motion for a fully vision-controlled instrument.

`Webcam` → `Image Analyze (Motion)` → `Synth Pitch/Filter`



CHAPTER

Nineteen

Output & Sharing

The final door out, to your speakers, your screen, another app, or a real projector.

Output

UTILITY · TERMINAL

The last node in every patch, sends your visuals and audio to the screen and speakers, and can record both to disk at once as an H.264 movie, ready to share or edit further.

TRY IT hit Record on your finished Layer Stack and audio mix to bounce a performance straight to file.

`Layer Stack (Visuals) + Audio Mix` → `Output`

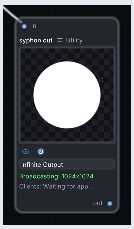
Syphon Out

UTILITY · IN

Broadcasts whatever image, 3D render, or shader is plugged into it out to other Mac apps in real time, Resolume, OBS, TouchDesigner, MadMapper, via zero-copy GPU memory sharing, with a customizable server name.

TRY IT send your generative render live into a VJ app like Resolume for a live show while still tweaking the patch in Infinite.

`3D Scene` → `Syphon Out` → `(external app)`



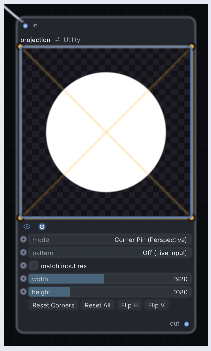
Projection

UTILITY · IN

A projection-mapping and warping node for driving real projectors, 4-corner perspective warping or a subdivided mesh- grid warp, with built-in alignment test patterns for lining up a physical projector.

TRY IT use the Grid test pattern to line up a projector on an odd-shaped surface, then switch to real content once corners are pinned.

`Rendered Scene` → `Projection` → `Syphon Out`



OSC Receive

UTILITY · OUT

Listens on a network port for OSC messages matching an address pattern and reports the last value received as a modulator, remapped to a low/high range, it patches in exactly like an LFO or Random node, just driven by another app or device instead.

TRY IT drive Bloom's amount from a TouchOSC fader on your phone for a wireless second pair of hands.

`OSC Receive` → `Mod Depth` → `Bloom (Amount)`

OSC Send

UTILITY · IN

The one node with no output at all: it reads whatever modulator is patched into it and sends its value out over the network as an OSC message, so a lighting console, Max/MSP patch, or another OSC-speaking app can react to what's happening inside Infinite.

TRY IT send Audio Analyze's bass band out over OSC to pulse a real DMX light rig on every beat.

`Audio Analyze (Bass)` → `OSC Send`

CHAPTER

Twenty

The Field Language, Programmable Nodes

The newest, deepest chapter in the palette: five nodes that don't offer a preset list at all, they offer a text box. Write a little math, and Infinite compiles it straight to the right engine for the job.

Five nodes let you write your own behavior directly into a node instead of picking from a fixed list. Type a small block of code, a kernel, and Infinite runs it automatically, once for everything it needs to touch: once per point on a mesh, once per pixel on screen, once per sample of audio. You never say how many times or how fast to run it, the node figures that out from what your code actually touches. It is a tiny programming language built for exactly one job, and it turns each of these five nodes into a blank canvas instead of a fixed preset list.

FieldPixel

SOURCE · IMAGE OUT

Runs your kernel once per pixel, compiled straight to a real GPU shader, an image generator or effect you write from scratch instead of picking off a shelf. It can declare its own state, a persistent texture it remembers between frames, so feedback-style effects that need memory are possible in one self-contained node.

TRY IT write a kernel that reads its own state and blends in the new frame for a self-contained feedback trail in one node.

`FieldPixel` → `Bloom` → `Output`

Field Modifier

3D · IN/OUT

Runs your kernel once per vertex of an incoming mesh, letting you rewrite position, normal, color, or any custom attribute with plain math instead of picking from a fixed deformer list. Leave the geometry input empty and it generates its own points instead, acting as a source.

TRY IT write P.y += bass to turn any mesh into a shape that visibly pulses with the low end.

`Geometry + Audio Analyze (Bass)` → `Field Modifier` → `Render 3D`

Field Primitive

3D · GEOMETRY OUT

A pure generator: builds 3D point geometry entirely from your kernel, starting from a topology scaffold, points, a plane, a sphere, a torus, that your code then reshapes. Presets like Fibonacci Sphere, Spiral, and Torus Knot show what's possible from just a formula.

TRY IT start from the Fibonacci Sphere preset and instance small spheres on the result for an evenly-scattered point cloud.

`Field Primitive` → `Instance on Points` → `Render 3D`

Field Synth

SYNTHS · AUDIO OUT

A polyphonic synthesizer whose entire voice algorithm is a kernel you write yourself, run per-voice on the audio thread, your own oscillator, FM algorithm, or wavetable scheme as plain math instead of a fixed engine.

TRY IT write a two-operator FM kernel from scratch and hear it respond to MIDI Notes like any other synth.

`MIDI Notes` → `Field Synth` → `Reverb`

Field Effect

AUDIOEFFECTS · IN/OUT

Runs a kernel once per audio sample as a custom effect, declare a bit of state for a delay line or filter memory, declare a param for an automatic modulatable knob, and write the DSP math yourself instead of hunting for the right built-in. Also cataloged in Chapter Twelve alongside the rest of the effects shelf.

TRY IT write a simple one-pole low-pass in a few lines and compare it against the built-in Audio Filter.

`Audio Source` → `Field Effect` → `Audio Out`

CHAPTER

Twenty-One

More Shapes & Slideshows

Ten more 2D sources: a picture slideshow that steps on the beat, and nine vector shapes that each do one thing well. Like Shape, every one is drawn from pure math, so they stay razor sharp at any size and every control can be animated.

Slideshow

SOURCE · IMAGE OUT

Steps through a folder of images on a beat interval, with an optional crossfade between them. Point it at a folder and it becomes a self-running photo reel locked to the tempo.

TRY IT set the interval to one bar and turn the crossfade up for a slow, dreamy gallery.

`Slideshow` → `Bloom` → `Output`

Ellipse

SOURCE · IMAGE OUT

A filled or stroked ellipse. Aspect stretches it away from a perfect circle, so one dial takes you from a dot to a wide lens shape.

TRY IT wire an LFO into Aspect so the circle squashes and stretches like a breathing pupil.

`LFO` → `Ellipse (Aspect)` → `Output`

Rectangle

SOURCE · IMAGE OUT

A filled or stroked rectangle with an optional corner radius. The simplest building block for panels, bars, frames, and mattes.

TRY IT use it as a hard-edged matte to cut a window out of a busy video.

`Rectangle` → `Set Alpha` → `Output`

Rounded Rect

SOURCE · IMAGE OUT

A rectangle with rounded corners. It is the same as Rectangle, but the corner radius gets its own control so you can animate it directly.

TRY IT modulate the corner radius from square to pill shape on every kick.

`Audio Analyze (Onset)` → `Rounded Rect (Radius)` → `Output`

Polygon

SOURCE · IMAGE OUT

A regular polygon, and Sides sets how many. Three is a triangle, five a pentagon, and a high number slides toward a circle.

TRY IT step Sides up with a Pattern modulator and watch the shape grow a corner at a time.

`Pattern` → `Polygon (Sides)` → `Output`

Hexagon

SOURCE · IMAGE OUT

A six-sided regular polygon. It is a Polygon with Sides locked at six, for when you simply want honeycomb geometry with one less thing to dial in.

TRY IT tile it with Mirror Tile for an instant honeycomb pattern.

`Hexagon` → `Mirror Tile` → `Output`

Crescent

SOURCE · IMAGE OUT

A crescent-moon shape, made by subtracting one circle from another. Move the two circles apart and the crescent fattens or thins.

TRY IT glow it up with Bloom and drop it over a dark gradient for a moon.

`Crescent` → `Bloom` → `Output`

Superellipse

SOURCE · IMAGE OUT

A squircle: a shape that lives between a rectangle and an ellipse. Inner Ratio tunes how square or how round it feels, the same silhouette you see on app icons.

TRY IT sweep Inner Ratio with an LFO to morph a circle into a rounded square and back.

`LFO` → `Superellipse (Inner Ratio)` → `Output`

Teardrop

SOURCE · IMAGE OUT

A teardrop or droplet outline: round at the bottom, pointed at the top. Handy for water, ink, and leaf-like motifs.

TRY IT repeat it in a ring with Kaleidoscope for a flower made of droplets.

`Teardrop` → `Kaleidoscope` → `Output`

Chevron

SOURCE · IMAGE OUT

An angled bracket, or chevron. Stack and repeat it for arrows, speed lines, and rank badges.

TRY IT feed it into Mirror Tile to build a scrolling zig-zag pattern.

Chevron → Mirror Tile → Output

CHAPTER

Twenty-Two

Alpha, Keys & Edges

Transparency tools and edge finders. Cut a subject out of its background, clean up the matte, and fix the dark or bright halos that show up when layers don't blend right. The last four nodes find edges and tile images seamlessly.

Show Alpha

COMPOSITING · IN/OUT

Shows the alpha channel as a plain greyscale picture: white is solid, black is transparent. It is a viewer, not an edit. It swaps the colour for the matte so you can see exactly what a key produced, and it is the first thing to reach for when a composite goes wrong.

TRY IT park it after a Chroma Key to see the matte and check for holes.

Video In → Chroma Key → Show Alpha → Output

Set Alpha

COMPOSITING · IN/OUT

Replaces the alpha channel outright with the brightness of a second image you patch in: a hand-drawn matte, a Shape, a Ramp, or another branch's Show Alpha. With nothing in the second input, the picture is forced fully opaque.

TRY IT patch a Ramp into the second input to fade a photo out from left to right.

Image Source → Set Alpha (Ramp) → Output

Alpha Invert

COMPOSITING · IN/OUT

Swaps transparent for solid and back, and leaves the colour alone. It is the invert-matte switch a key would otherwise need a second node to get.

TRY IT invert a Luma Key to keep the dark parts of a shot instead of the bright ones.

Video → Luma Key → Alpha Invert → Output

Alpha from Luma

COMPOSITING · IN/OUT

Builds a new alpha channel from the picture's own brightness, so a black background becomes see-through with nothing to key against. Invert flips which end is solid, for a white-on-black source.

TRY IT run fire or smoke footage through it to lift the flames off their black backdrop.

Video → Alpha from Luma → Blend

Alpha Levels

COMPOSITING · IN/OUT

The Levels remap, applied to the alpha channel only. Low and High clamp and stretch the matte's range (choking a soft edge tighter, or spreading it wider) and Gamma bends the falloff in between. The go-to cleanup pass after Chroma Key or Alpha from Luma.

TRY IT raise Low a touch after a key to remove the faint haze left around a subject.

`Chroma Key` → `Alpha Levels` → `Output`

Premultiply

COMPOSITING · IN/OUT

Converts between premultiplied alpha (colour already scaled by its own alpha) and straight alpha. Only reach for it when edges look wrong: a dark fringe on a composite usually means straight alpha needs premultiplying, and a bright halo means the reverse.

TRY IT if a keyed subject has a dark outline over a bright background, add this before the Blend.

`Chroma Key` → `Premultiply` → `Blend`

Chroma Key

COMPOSITING · IN/OUT

Removes a chosen key colour, green-screen or blue-screen style, matching it in a way that ignores brightness. Tolerance and Softness shape the edge, Spill Removal tames colour-cast fringes, and the show option previews the matte instead of the keyed picture.

TRY IT pick the green of a backdrop, then raise Spill Removal until the green glow on hair disappears.

`Video In` → `Chroma Key` → `Blend` → `Output`

Luma Key

COMPOSITING · IN/OUT

Keys out a range of brightness (Low to High) instead of a colour, with Softness at the edges and Invert to flip which range is kept. The show option previews the matte.

TRY IT key out a white sky from a skyline photo and drop a new sky behind it.

`Image Source` → `Luma Key` → `Blend` → `Output`

Mirror Tile

EFFECTS · IN/OUT

Tiles the image and flips every other tile so the edges meet seamlessly, unlike a plain repeat. Tiles sets how many repeats you get.

TRY IT tile a Noise source to make a seamless texture for a 3D material.

`Noise` → `Mirror Tile` → `Material`

Normal Map

EFFECTS · IN/OUT

Works out a tangent-space normal map by treating the picture's brightness as a height field. The output switch can also show the raw normals, the source height, or the steepness of the slope.

TRY IT turn a Noise or Texture into a bump map for a Material's surface.

`Texture (Brick)` → `Normal Map` → `Material`

Edge Sobel

EFFECTS · IN/OUT

Classic Sobel edge detection. It outputs how sharply the picture changes as greyscale lines, and Invert flips black and white.

TRY IT wire it into a Displacement node so only the edges of a photo lift off the surface.

`Image Source` → `Edge Sobel` → `Output`

Edge Outline

EFFECTS · IN/OUT

Finds edges from the brightness gradient and draws them as flat-colour outlines over the original picture. Thickness and Threshold control how bold and how many.

TRY IT give a video a hand-inked comic look by adding a black outline over it.

`Video` → `Edge Outline` → `Output`

CHAPTER

Twenty-Three

More 3D Solids & Mesh Surgery

Thirteen more 3D primitives (platonic solids, twisted surfaces, gears and arrows), a depth-map unprojector, and the mesh operators: turn a surface into points, thicken it, wireframe it, blow it apart, or delete part of it. Every one takes geometry in and gives geometry out, so they chain like any other 3D node.

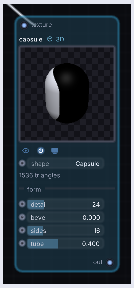
Capsule

3D · GEOMETRY OUT

A cylinder with half-sphere caps, like a pill. The tube control sets the radius of the caps and the body together.

TRY IT stretch a Capsule and spin it for a quick pill, bone, or lightsaber shape.

`Capsule` → `Material` → `Render 3D`



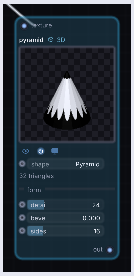
Pyramid

3D · GEOMETRY OUT

A pyramid. Sides sets how many sides the base has, so 4 is the classic square pyramid and larger numbers lean toward a cone.

TRY IT scatter a handful with Instance on Points for a desert of pyramids.

`Pyramid` → `Instance on Points` → `Render 3D`



Prism

3D · GEOMETRY OUT

A prism: a polygon extruded into a solid. Sides sets the polygon, so 3 is a triangular prism and 6 is a hex bar.

TRY IT stack several prisms of different sizes for a skyline.

`Prism` → `Array` → `Render 3D`

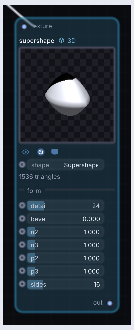
Supershape

3D · GEOMETRY OUT

A shape built from the superformula. Depth, tooth depth and hub hole warp it into gear-like or organic silhouettes, from starfish to flowers to cogs.

TRY IT sweep tooth depth with an LFO and the shape slowly morphs from smooth to spiky.

LFO → Supershape (Tooth Depth) → Render 3D



Tetrahedron

3D · GEOMETRY OUT

A four-faced platonic solid: the simplest 3D shape that can enclose space, made of four triangles.

TRY IT put one under a Wireframe node for a crisp geometric diagram look.

Tetrahedron → Wireframe → Render 3D

Octahedron

3D · GEOMETRY OUT

An eight-faced platonic solid: two pyramids joined at the base, like a diamond.

TRY IT give it a glassy Material and a slow spin for a floating crystal.

Octahedron → Material → Render 3D

Dodecahedron

3D · GEOMETRY OUT

A twelve-faced platonic solid with pentagon faces. It is the rounder of the classic solids and reads almost like a ball.

TRY IT run it through Explode to blow the twelve faces apart like a petal.

Dodecahedron → Explode → Render 3D

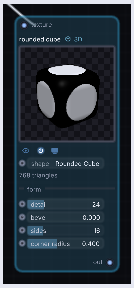
Rounded Cube

3D · GEOMETRY OUT

A cube with soft edges and corners. Corner radius sets how rounded it gets, from sharp-edged to almost a sphere.

TRY IT animate the corner radius on a beat for a bouncy, squishy cube.

Rounded Cube → Material → Render 3D



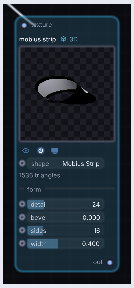
Mobius Strip

3D · GEOMETRY OUT

A twisted strip with one side and one edge. Width sets how wide the band is.

TRY IT add Solidify so the one-sided ribbon becomes a solid band you can light properly.

Mobius Strip → Solidify → Render 3D



Klein Bottle

3D · GEOMETRY OUT

The classic bottle with no inside or outside. It is a closed surface that passes through itself, which makes it a striking thing to light and rotate.

TRY IT give it a slow spin and a bright rim light for an impossible-object hero shot.

Klein Bottle → Material → Render 3D

Gear 3D

3D · GEOMETRY OUT

A 3D cogwheel. Depth sets the thickness, and tooth depth and hub hole shape the teeth and the centre bore. It carries the 3D suffix because the 2D Shape node already has its own Gear.

TRY IT rotate two Gear 3Ds in opposite directions for a clockwork look.

Gear 3D → Material → Render 3D

Star 3D

3D · GEOMETRY OUT

A 3D star extruded into a solid. Depth sets how thick it is, and inner ratio sets how deep the points cut in. It carries the 3D suffix because the 2D Shape node already has its own Star.

TRY IT instance a few hundred with random spin for a shower of stars.

Star 3D → Instance on Points → Render 3D

Arrow 3D

3D · GEOMETRY OUT

A 3D arrow made of a shaft and an arrowhead. It carries the 3D suffix because the 2D Shape node already has its own Arrow.

TRY IT scatter arrows across a surface and aim them for a flow-field display.

Arrow 3D → Instance on Points → Render 3D

Depth Projection

3D · GEOMETRY OUT

Turns a depth map, plus an optional colour image, into a 3D point cloud or a triangulated surface. It supports pinhole camera, planar, radial and cylindrical projections, depth curves, and edge tearing.

TRY IT feed a depth map and its matching photo to pop a flat picture into a shallow 3D scene.

Image Source (Depth) → Depth Projection → Render 3D

Mesh to Points

3D · GEOMETRY OUT

Samples a mesh's vertices as a point cloud, ready for Instance on Points or Metaballs.

TRY IT turn a model into points and put a tiny sphere on each one.

Mesh to Points → Instance on Points → Render 3D

Mesh to Edges

3D · GEOMETRY OUT

Samples points along a mesh's edges as a point cloud, for Instance on Points or Metaballs. A good way to trace the outline of a shape with beads.

TRY IT instance small spheres along the edges of a cube for a bead-wire frame.

Mesh to Edges → Instance on Points → Render 3D

Mesh to Faces

3D · GEOMETRY OUT

Samples the centre point of every face as a point cloud, for Instance on Points or Metaballs. One point per face, so the density follows the mesh.

TRY IT stand a small object on every face of a dodecahedron.

Mesh to Faces → Instance on Points → Render 3D

Solidify

3D · GEOMETRY OUT

Gives a flat or open surface real thickness, turning it into a solid shell. Keep original also keeps the source surface.

TRY IT thicken a Plane or a Mobius Strip so it has real edges to catch the light.

Mobius Strip → Solidify → Render 3D

Wireframe

3D · GEOMETRY OUT

Replaces solid faces with a thin tube along every edge, at the Thickness you choose. A solid becomes a cage.

TRY IT wireframe a Dodecahedron and add Bloom for a neon-sign look.

Dodecahedron → Wireframe → Render 3D

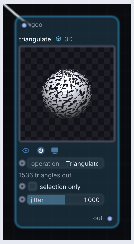
Triangulate

3D · GEOMETRY OUT

Re-cuts the mesh into triangles. Jitter randomises how each face is split, giving a more organic, less uniform look.

TRY IT triangulate with some Jitter, then run Explode for a shattered low-poly feel.

Sphere → Triangulate → Explode



Normals

3D · GEOMETRY OUT

Recomputes how the surface faces the light. Flat shade gives hard per-face edges instead of smooth shading, and flip reverses which way every face points.

TRY IT switch on flat shade to turn a smooth sphere into a faceted gem.

Sphere → Normals → Render 3D

Explode

3D · GEOMETRY OUT

Pushes every face outward from the mesh's centre along its own direction, by Amount, with a different amount per face set by Seed.

TRY IT wire an envelope into Amount so the shape bursts apart on each hit.

Envelope → Explode (Amount) → Render 3D

Screw

3D · GEOMETRY OUT

Sweeps a profile around an axis in a spiral. Steps sets the smoothness, Turns the number of revolutions, Rise per turn the pitch, and Radius pushes the sweep outward.

TRY IT sweep a small circle to make a spring, or a flat strip for a spiral staircase.

Shape → Screw → Render 3D

Delete

3D · GEOMETRY OUT

Removes the faces marked by an upstream Select node (or the instances, when it sits after Instance on Points). Turn on keep selected instead to flip it: everything except the selection goes, and only the selection stays.

TRY IT select the top half of a sphere and delete it to make a bowl.

Sphere → Select → Delete → Render 3D

Set Vertex Color

3D · GEOMETRY OUT

Writes colour onto each element: vertex colour on a mesh, or per-particle colour on a point cloud, whichever the input carries. Source picks where the colour comes from: one flat colour, a two-colour ramp, random per element, a texture sampled at each element's UV, or the swatches of a Palette node. Render 3D reads it without being told.

TRY IT colour a scatter of points from a Palette so every point picks one swatch.

Mesh to Points → Set Vertex Color (Palette) → Render 3D

CHAPTER

Twenty-Four

Loops, Learning & Listening

A live audio looper, a 16-pad sample player, three new audio effects, a level meter, two modulator helpers, and the Prediction family: nodes that listen to what you play, learn your habits, and carry on in the same style. The learning nodes keep what they learn (some of it is saved with the patch), so a good take can be reused.

Looper

SYNTHS · AUDIO OUT

A live audio looper. Wire in the sound, press Rec, and the take ends on its own after the take length and then loops. Play restarts it, Dub layers new sound over it, and Clear empties it. Take length can be a musical division, or free. Every take is shifted by your audio interface's measured latency, so a loop played in time sits on the grid. The loop is kept in memory (up to 60 seconds) and is not saved with the patch.

TRY IT record a one-bar drum pattern, then Dub a bassline on top of it.

Audio File → Looper → Audio Out

MPC

SYNTHS · AUDIO OUT

A 16-pad sample player in the spirit of an MPC. Each pad holds one sample and one play mode: One shot plays the whole sample, Gate plays while the pad is held, and Loop toggles a looping playback on and off. Hit pads with the mouse, with a MIDI controller, or with notes 36 to 51 on the note input, the usual pad-controller layout. Every pad is its own voice with its own volume, pitch, pan, speed, fine tune and fades, and a pad can be Synced to a musical division so a hit waits for the next grid line. All pads mix to one stereo output.

TRY IT load a kick, snare and hat on three pads and play them from a MIDI pad controller through a MIDI Notes node.

MIDI Notes → MPC → Audio Out

Resonator Bank

AUDIOEFFECTS · IN/OUT

Up to 16 band-pass resonators running side by side, forming a tuned bank of ringing filters, including a metallic tuning mode. Feed it noise or drums and it rings like a struck bell or plate.

TRY IT push a drum loop through it in metallic mode for a gamelan-style shimmer.

Audio File → Resonator Bank → Audio Out

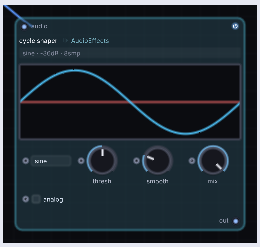
Cycle Shaper

AUDIOEFFECTS · IN/OUT

Distorts and crossfades the shape of a single waveform cycle. A way to bend a plain tone into harder, brighter, or stranger timbres.

TRY IT run a plain sine through it and sweep the shaping for a rubbery, evolving lead.

Oscillator → Cycle Shaper → Audio Out



Spec Blur

AUDIOEFFECTS · IN/OUT

A streaming phase vocoder that smears the spectrum over time. Sharp hits dissolve into a harmonic cloud, with tilt, phase diffusion and freeze controls. It adds 42.7 ms of delay, and its mix defaults to fully wet.

TRY IT freeze a held chord and let it hang as a pad while you play over the top.

Wavetable → Spec Blur → Reverb → Audio Out

Audio Meter

UTILITY · IN/OUT

A stereo level meter with separate left and right bars on a -60 to +3 dBFS scale. Each bar shows the average level (solid) inside the peaks (faint), with a peak-hold line and the loudest peak as a number on top that turns red at 0 dBFS. Audio passes through unchanged, and it measures whatever it is patched to even with its output unconnected, so it works as a tap on any cable.

TRY IT hang one off the master cable to watch your headroom as you mix.

Audio File → Audio Meter → Audio Out

Velocity to CV

MODULATORS · IN/OUT

Turns the velocity of incoming notes into a 0 to 1 modulator, with an adjustable velocity range. Hit harder and the value goes higher.

TRY IT let note velocity control the Bloom amount, so loud notes flare.

MIDI Notes → Velocity to CV → Bloom (Amount)

CV Recorder

MODULATORS · IN/OUT

Records any modulator you patch in. Press Rec, perform a gesture, and it loops the take back with adjustable speed and low and high range. Takes are timed in beats, follow the tempo, pause with the transport, and are saved with the patch.

TRY IT record a hand-played XY Motion Pad swirl, then loop it into Twirl.

XY Motion Pad → CV Recorder → Twirl (Angle)

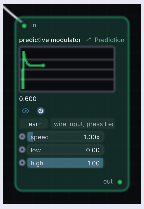
Predictive Modulator

PREDICTION · IN/OUT

Learns a small model of one modulator you patch in, then keeps that motion going on its own once learning stops. Draw a gesture once and it carries on in the same spirit.

TRY IT learn a few seconds of a knob move, stop, and it keeps wandering in the same way.

Macro Knob → Predictive Modulator → Bloom (Amount)



Predictive Coloring

PREDICTION · IN/OUT

Learns what graded footage looks like to you and grades incoming frames toward it. Press Learn on footage you've already graded to build a target look, or leave self normalize on to auto-level without one. Mix blends the grade in, and Wander makes it drift gently among looks it has actually seen.

TRY IT learn from a graded still, then run raw video through it to match.

Video In → Predictive Coloring → Output

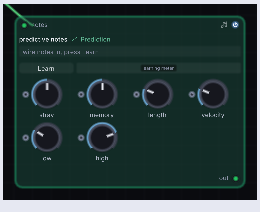
Predictive Notes

PREDICTION · NOTE OUT

Wire a note chain in and press Learn. It listens (passing the notes through), learns pitches, rhythm, lengths and velocities, then plays on its own in that style. Stray at the bottom replays the phrase, the middle plays in character, and the top ignores what it learned and picks freely. What it learned is saved with the patch.

TRY IT play a short phrase, press Learn, and let it improvise a variation.

MIDI Notes → Predictive Notes → Wavetable



Predictive Quantize

PREDICTION · NOTE OUT

A groove quantizer instead of a grid one. It listens to the real spacing between your notes and corrects them in the same pass, with no Learn button, adapting the more you play. Its learned feel is shared by every Predictive Quantize in every patch and keeps only your most recent notes.

TRY IT tighten a loose keyboard take without making it sound robotic.

MIDI Notes → Predictive Quantize → Wavetable

Predictive Velocity

PREDICTION · NOTE OUT

A learned dynamics curve instead of a hand-drawn one. It listens to how hard you play and remaps velocities in the same pass, with no Learn button, adapting as you go. The curve is shared across every patch and keeps only your most recent notes.

TRY IT even out a lumpy performance while keeping your own touch.

MIDI Notes → Predictive Velocity → Wavetable

Predictive Rhythm

PREDICTION · NOTE OUT

Works like Predictive Notes, but learns pitch as a distance from a root note instead of absolute pitches. What comes back is a repeating pattern anchored on one note with the occasional deviation, such as C2 C2 C2 D3 C2 C2 C2. Root is found automatically, and changing it afterwards transposes the whole pattern without relearning.

TRY IT play a bass riff, Learn it, then change Root to move the pattern up a fifth.

MIDI Notes → Predictive Rhythm → Wavetable

CHAPTER

Twenty-Five

Sketch, Field Notes & Signals

Code-drawn images and meshes, a Field script that plays notes, a MIDI file player and out, and the three Tracking nodes that turn a camera into modulators. Sketch and Sketch 3D turn every param() call into a knob you can modulate, like any other control.

Sketch

SOURCE · IMAGE OUT

Draws an image from a JavaScript sketch you write in the node, with loops, arrays and recursion and a p5.js-style API (rect, circle, line, beginShape, push/pop, translate, rotate, text). A mistake shows its line number and the last working picture stays up. Drop an .svg on the canvas to animate it by id.

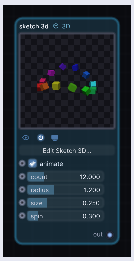
TRY IT write one param("size", 50, 5, 200) call and wire an LFO into the knob it creates.

`LFO` → `Sketch (size)` → `Output`

Sketch 3D

3D · GEOMETRY OUT

Builds 3D geometry from a JavaScript sketch: box, sphere, cylinder, cone, torus, tube and beginShape/vertex/endShape, moved with translate, rotate and scale. It is the loop-and-recursion counterpart of Field Primitive, and the result patches into Render 3D like any other geometry.



TRY IT stack a loop of rotated boxes into a spiral staircase and modulate the twist.

`Sketch 3D` → `Render 3D` → `Output`

Field Notes

NOTES · NOTE OUT

A note node whose body is a Field script. It runs per sample against the transport beat, and note(pitch, vel, len) inside an if sends a note out. tick(div) fires on each beat division, deg(d) is a scale degree under Root and Scale, and rand() is 0 to 1.

TRY IT fire a random scale degree on every eighth note and send it to a synth.

`Field Notes` → `Wavetable` → `Audio Out`

MIDI File

NOTES · NOTE OUT

Plays a Standard MIDI File as a note source. It follows the transport, and the project tempo rules, so the file's own tempo is ignored. Transpose, position, speed, velocity and loop are all controls, and drum-channel notes are never transposed.

TRY IT drop a .mid on the canvas and send it to an arpeggiator for a new take.

MIDI File → Arpeggiator → Wavetable

MIDI Out

UTILITY · NOTE IN

Plays a hardware or virtual MIDI synth from a note cable on one channel. Four CC rows send a controller number and a modulatable value only when it changes, an optional clock makes Infinite the master clock, and an offset lines up the synth's latency. Anything still held is released on delete, bypass or panic.

TRY IT patch an LFO into a CC row to sweep a filter on an external synth.

Note Sequencer → MIDI Out

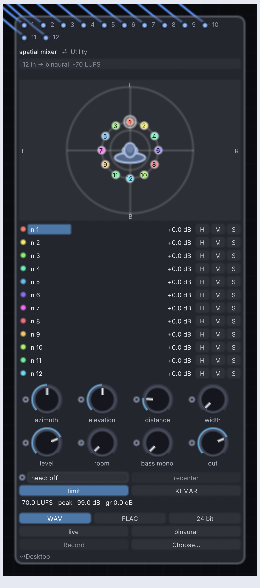
Spatial Mixer

UTILITY · AUDIO IN

Mixes audio in 3D for headphones and plays it straight to the output. Each input is a dot around a head, and its angle, distance and height change how it is heard. A limiter holds -1 dB, it records its own WAV or FLAC, and head tracking works with AirPods on macOS 14 and later.

TRY IT put a bass on the left, a lead on the right and a pad above your head.

Wavetable → Spatial Mixer



Hand Track

MODULATORS · IMAGE IN

Tracks one hand from an image or camera wire into modulators: palm and fingertip position, pinch, open, roll and size, plus six gestures (fist, point, peace, thumbs up, ok, rock). Needs the Tracking pack from Settings > Extensions.

TRY IT map pinch to a bloom amount and wave your hand to play the picture.

`Video In` → `Hand Track` → `Bloom (Amount)`

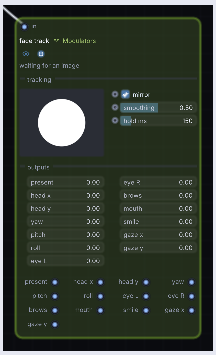
Face Track

MODULATORS · IMAGE IN

Tracks one face into modulators: head position, turn, nod and tilt, each eye open or closed, brows, mouth open, smile, and where the eyes look. Needs the Tracking pack from Settings > Extensions.

TRY IT open your mouth to open a filter cutoff.

`Video In` → `Face Track` → `Audio Filter (Cutoff)`



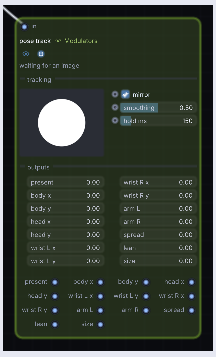
Pose Track

MODULATORS · IMAGE IN

Tracks one person's body into modulators: head, both wrists, how high each arm is raised, how far apart the wrists are, lean and distance. Keep the upper body in view. Needs the Tracking pack from Settings > Extensions.

TRY IT raise an arm to bring a chord in.

`Video In` → `Pose Track` → `Gain (Level)`



CHAPTER

Twenty-Six

Meshes, Curves & Dither

Newer geometry operators for turning points into surfaces and cells, cleaning up curves, and a dither filter that keeps gradients smooth.

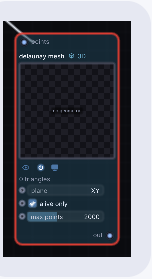
Delaunay Mesh

3D · GEOMETRY OUT

Point cloud in, triangulated surface out. Points are projected onto the chosen plane, joined into Delaunay triangles, and lifted back to their real 3D positions, so a scatter with height becomes a terrain. A mesh is flagged red and ignored, so use Mesh to Points first. Max points thins very large clouds.

TRY IT turn a shape into points and wire them in for an instant low-poly surface.

Sphere → Mesh to Points → Delaunay Mesh



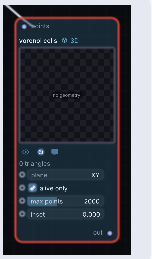
Voronoi Cells

3D · GEOMETRY OUT

Point cloud in, one polygon cell per point out, for cracked-glass and fracture looks. Every cell is the region closer to its point than to any other, and hull points are skipped. Inset pulls each cell toward its point to leave gaps.

TRY IT raise inset to split a flat sheet into glass shards.

Sphere → Mesh to Points → Voronoi Cells



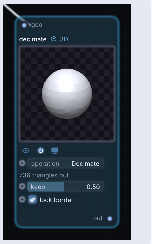
Decimate

3D · GEOMETRY OUT

Reduces triangle count while keeping the overall shape. Keep is the fraction of triangles to leave, and Lock border holds open edges in place. UVs, colours and hard shading survive, and very low settings stop when no safe collapse remains.

TRY IT drop keep to 10% for a low-poly look on a dense sphere.

Sphere → Decimate → Render 3D



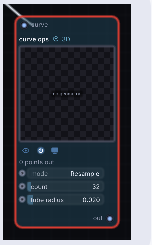
Curve Ops

3D · GEOMETRY OUT

Curve in, curve out. Resample spaces points evenly, Simplify drops points that barely change the shape, Smooth relaxes corners while the ends stay put, and Offset draws a parallel curve at a signed distance. The result can still drive a Path.

TRY IT offset a hand-drawn curve to draw a contour line around it.

Curve Ops → Path



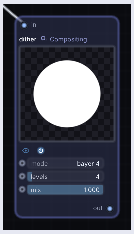
Dither

COMPOSITING · IMAGE IN/OUT

Reduces the image to a few levels per channel like Posterize, but spreads the rounding error as a fixed pattern, so gradients keep their tone instead of banding. Mode picks an ordered Bayer matrix (2, 4 or 8) or a noise pattern, and Mix blends back toward the original.

TRY IT set three levels with an 8x8 matrix for a retro print look.

`Video` → `Dither` → `Output`



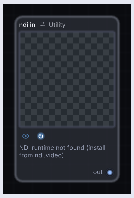
NDI In

UTILITY · IMAGE OUT

Receives any NDI video source on the network as an image. Pick it from the list, which rescans in the background. Video only. Needs the free NDI runtime from ndi.video.

TRY IT pull a second computer's screen in and feed it through effects.

`NDI In` → `Bloom` → `Output`



NDI Out

UTILITY · IMAGE IN

Publishes the connected image as a named NDI video source on the local network for OBS, vMix or Resolume. Video only. Without the NDI runtime the node says so and everything else keeps working.

TRY IT send your finished patch to OBS with no capture card.

`Video` → `NDI Out`

CHAPTER

Twenty-Seven

New Instruments, Effects & Modulators

The sound side of v0.6: four new synths, a designed-room reverb, a three-stage saturator, spectral tools, and modulators that give every note its own value.

FM

AUDIO · SYNTH

A 4-operator FM synth: four sine operators wired by one of 11 algorithms, from a 4>3>2>1 stack to four carriers side by side. Each operator follows the note at a ratio or sits at a fixed frequency, and has its own level and envelope. Feedback lets operator 4 modulate itself, from a brighter sine to a buzz.

TRY IT pick an algorithm with two stacked pairs and raise operator 2's level for a bell.

MIDI Notes → FM → Audio Out

String

AUDIO · SYNTH

A physically modelled string: every note is a waveguide string tuned exactly to its pitch, on a shared wooden body. The exciter plucks it, bows it (the bow keeps it sounding while the note is held) or strikes it with a felt hammer that gets brighter the harder you hit. Position is where along the string it is excited.

TRY IT switch the exciter to bow and hold a chord for a sustained cello pad.

MIDI Notes → String → Audio Out

Partials

AUDIO · SYNTH

Playable resynthesis: a loaded sample is analysed once into tracked partials plus the noise they leave behind, then every note plays it back additively at the note's pitch. Transposing keeps the sample's formants where they were, so a voice an octave up still sounds like the same voice. Partials thins the stack from the top and morph crossfades between sets.

TRY IT drop in a voice sample and play it as a keyboard that never turns into a chipmunk.

MIDI Notes → Partials → Audio Out

Analog

AUDIO · SYNTH

A polyphonic virtual-analog voice: two oscillators (the first a unison stack), a sub oscillator, noise, a drive stage, a nonlinear filter (Moog ladder, MS-20, diode ladder or SVF) and an amplitude envelope across up to 8 voices. Narrow-band oscillators and a 4x drive and filter keep aliasing under -80 dB, and default patches are in tune.

TRY IT stack three detuned voices, open the MS-20 filter and sweep the cutoff with an envelope.

MIDI Notes → Analog → Audio Out

Saturator

AUDIO · EFFECT

Three saturation stages, each with its own curve (tube, tape, diode, fold, rectify, crush) and amount. Routing wires them serial, parallel, mid-side, 3-band or feedback. It runs at 8x with anti-aliased curves and ships with 8 presets.

TRY IT use 3-band routing: gentle tape on the lows, fold on the highs.

Analog → Saturator → Audio Out

Convolution Reverb

AUDIO · EFFECT

A reverb whose room you design: shape (room, hall, chamber, tunnel), size, material, distance, decay time and stereo spread. The node builds that room's impulse response in the background and shows it. Load file uses a recorded impulse response instead (up to 10 s). It adds no latency.

TRY IT make a long tunnel with a hard material for a metallic, echoing space.

String → Convolution Reverb → Audio Out

Key-Snap

AUDIO · EFFECT

Moves every spectral peak of the sound to the nearest note of a scale, so chords and layered sources stay in key. Snap blends between untouched and fully in tune, glide sets how fast a peak slides to its new note, and global key follows the transport. Latency is 42.7 ms.

TRY IT run a detuned pad through it and set the scale to minor.

Partials → Key-Snap → Audio Out

Spectrum Slide

AUDIO · EFFECT

Morphs the sound into the one on the 'to' pin by sliding its spectral peaks toward the other's positions instead of cross-fading, so a 440 Hz tone becomes 880 Hz by passing through 660 Hz. Latency is 42.7 ms.

TRY IT automate Slide from 0 to 1 over a bar to turn a bell into a voice.

String → Spectrum Slide → Audio Out

Shape Resonator

AUDIO · EFFECT

Rings the vibration modes of the shape on its 'shape' pin, like striking that object. Tune sets the lowest mode's pitch; the others follow from the geometry, so a flat plate sounds bell-like and a long bar nearly harmonic. Pos is where it is struck.

TRY IT feed it a 3D shape and morph the shape to change the timbre.

Sphere → Shape Resonator → Audio Out

MSEG

MODULATOR · SIGNAL OUT

A shape you draw, played over time. Click empty space to add a point, drag to move it, right-click to remove it, and bend a segment with the ring on it. Rate sets how long one pass takes; a trig cable starts the shape on each rising edge. A per-voice switch gives each note its own copy.

TRY IT draw a rising-then-falling curve and cable it into a filter cutoff.

MSEG → Analog (Cutoff)

Voice Spread

MODULATOR · SIGNAL OUT

A different value for each note of a chord. Cable it into an Analog or Wavetable knob and every note gets its own setting between low and high, by its place among the held keys. Linear spaces them evenly, golden ratio scatters them, primes spaces them unevenly.

TRY IT spread pan across a chord so the notes fan out across the stereo field.

Voice Spread → Analog (Pan)

Note Expression

MODULATOR · SIGNAL OUT

Each note's own MPE pressure or slide. Turn on MPE on MIDI Notes, then cable this into an Analog or Wavetable knob: every voice follows the pressure or slide of the finger playing it, smoothed over 5 ms.

TRY IT cable pressure into filter cutoff and play with an MPE controller.

`MIDI Notes (MPE)` → `Note Expression` → `Analog (Cutoff)`

Drift

MODULATOR · SIGNAL OUT

Learns where you leave each knob and how fast you move it, then keeps the knob going after you let go: it settles into your usual places and wanders between them at your pace. Speed, Stray and Momentum shape the wandering, and Link lets your live hand activity stir it.

TRY IT play for a minute, let go, and watch the patch keep evolving.

`Drift` → `Analog (Cutoff)`

CHAPTER

Twenty-Eight

New Compositing, 3D & Mesh Tools

The visual side of v0.6: motion analysis, fluid and fog, physics, plants from text, and the primitives and mesh operators earlier editions of this guide missed.

Optical Flow

COMPOSITING · IMAGE IN/OUT

How far each part of the picture moved since the last frame, as a vector per pixel (red and green carry x and y, blue is speed). Pyramidal Lucas-Kanade on the GPU. View Colour shows direction as hue and speed as brightness.

TRY IT displace a second picture along the motion of a video.

Video → Optical Flow → Displace

Time Machine

COMPOSITING · IMAGE IN/OUT

Shows each pixel from a different moment: it keeps the last frames and a grey map (the second input) says how far back every pixel looks. A left-to-right ramp, the default, gives a slit-scan. Length sets how many frames are kept, up to 60.

TRY IT feed it a radial gradient so the middle of the picture lags behind the edges.

Video → Time Machine → Output

Fluid

COMPOSITING · IMAGE IN/OUT

GPU stable fluids with multigrid pressure, vorticity, obstacles, a force input and a built-in stirrer. Ink or any image is carried along by the flow.

TRY IT use the Optical Flow of a dancer as the force input and let the ink follow them.

Video → Optical Flow → Fluid

Volume

COMPOSITING · IMAGE IN/OUT

Ray-marched fog, smoke and cloud over an image, using the Render 3D camera and depth, with its own camera when none is connected. Long controls fold behind show more.

TRY IT put fog behind a lit 3D scene for depth.

Render 3D → Volume → Output

Rigid Bodies

3D · GEOMETRY OUT

Drops every separate piece of the input mesh as a rigid body (box, sphere or convex-hull collider) that falls, collides and stacks on a ground plane and optional walls. Mass, friction, bounce, damping, gravity, spin and a timed kick. The same patch plays back identically every time.

TRY IT scatter a field of cubes and press Reset to drop them again.

Cube → Rigid Bodies → Render 3D

L-System

3D · GEOMETRY OUT

Grows a plant or fractal from text. Type an axiom and rewrite rules, set the angle and the number of generations, and a 3D turtle draws it as a tube mesh and a followable curve.

TRY IT start with the rule X=F+[[X]-X]-F[-FX]+X;F=FF for a fern.

L-System → Render 3D

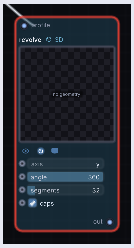
Revolve

3D · GEOMETRY OUT

Spins a profile curve around the X, Y or Z axis to make a surface of revolution: vases, bowls, rings, columns. A closed profile turned all the way round is a solid. Segments sets how smooth the turn is.

TRY IT draw half a vase outline with a Curve and revolve it.

Curve → Revolve → Render 3D



Proximity Lines

3D · GEOMETRY OUT

Joins points closer than Radius with thin lines, for plexus and constellation looks. Point cloud in, mesh of lines out. Max per point and Max lines keep a dense cloud from turning solid.

TRY IT scatter points with a Particle System and join the near ones.

Particle System → Proximity Lines → Render 3D

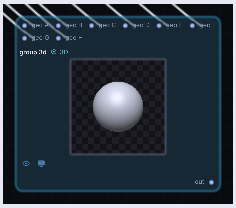
Group 3D

3D · GEOMETRY IN/OUT

Carries up to eight geometry inputs through one pin into Render 3D, so a scene is not limited to four geometry pins. Each input keeps its own material, texture and transform.

TRY IT gather a whole scene into one cable and send it to Render 3D.

Sphere → Group 3D → Render 3D



Subdivide

3D · GEOMETRY OUT

Subdivision surface. Each level roughly quadruples the triangle count, so higher levels get expensive fast. Smooth controls how much it rounds corners.

TRY IT one level on a cube gives a soft pillow.

Cube → Subdivide → Render 3D

Bevel

3D · GEOMETRY OUT

Rounds or chamfers every sharp edge above the sharp angle. Width is the bevel size; one segment gives a flat chamfer and more give a smooth round.

TRY IT bevel a box so its edges catch highlights.

Cube → Bevel → Render 3D

Extrude

3D · GEOMETRY OUT

Extrudes every face of the input mesh along its own normal by Distance, with Inset shrinking each face first.

TRY IT inset and extrude a sphere for spiky studs.

Sphere → Extrude → Render 3D

Twist

3D · GEOMETRY OUT

Twists the mesh around a chosen axis by Angle: the further a point is along the axis, the more it rotates.

TRY IT animate Angle with an LFO to wring a tall box like a towel.

Cube → Twist → Render 3D

Cone

3D · GEOMETRY OUT

A cone. Sides sets how many-sided it is.

TRY IT set Sides to 4 for a pyramid.

Cone → Render 3D

Cylinder

3D · GEOMETRY OUT

A cylinder. Sides sets how many-sided it is; low values give a prism.

TRY IT three sides gives a triangular prism.

Cylinder → Render 3D

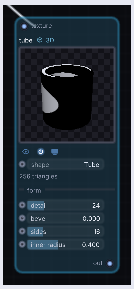
Tube

3D · GEOMETRY OUT

A hollow cylinder. Its inner radius sets the thickness of the wall.

TRY IT make a ring or a pipe.

Tube → Render 3D



Disc

3D · GEOMETRY OUT

A flat filled circle mesh, as opposed to Cylinder, which has height.

TRY IT use it as a floor or a coin.

Disc → Render 3D

Blob

3D · GEOMETRY OUT

An organic, wobbly rounded shape, randomised by seed-like controls rather than a precise geometric primitive.

TRY IT change the seed for a new pebble.

Blob → Render 3D

Heart

3D · GEOMETRY OUT

A heart-shaped outline.

TRY IT extrude it for a solid heart.

Heart → Extrude → Render 3D

Cross

3D · GEOMETRY OUT

A plus-sign shape.

TRY IT extrude it for a medical cross.

Cross → Extrude → Render 3D

Pie

3D · GEOMETRY OUT

A circular wedge. Inner Ratio controls how much of the circle is cut away.

TRY IT animate it into a loading ring.

Pie → Render 3D

CHAPTER

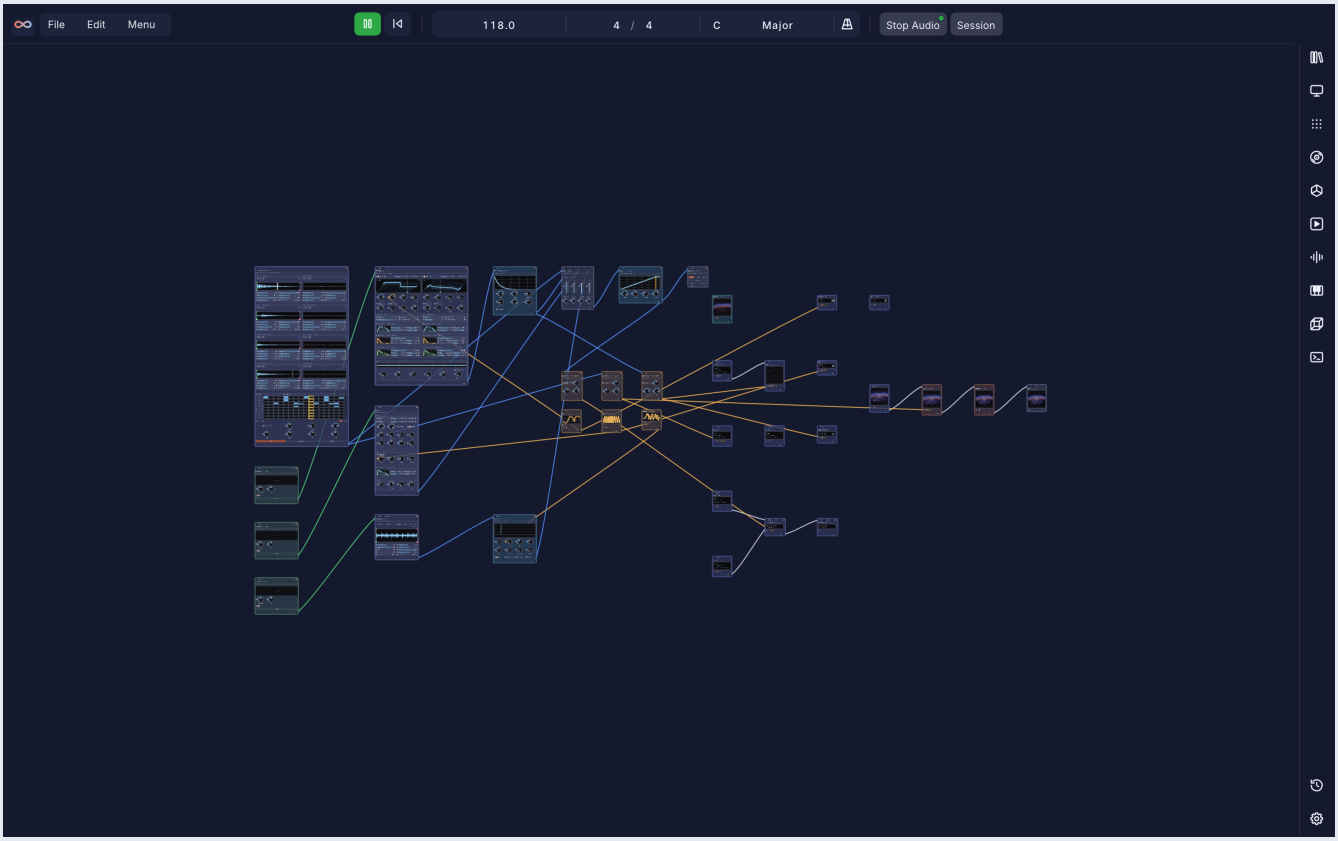
Twenty-nine

Using the App: Panels, Menus & Settings

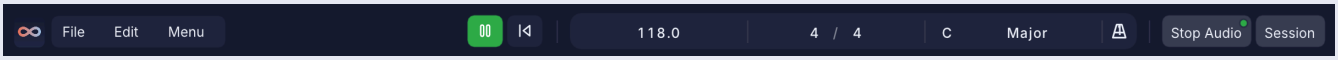
Everything around the nodes: the panels on the right, the strips along the bottom, the right-click menus, the dropdowns and the settings. Every picture here is a real screenshot of the app, taken by a script.

These are the questions people ask most: what does the library do, what is the modulation matrix, how do I build a performance page, how do clips and tracks work, what is in the right-click menu. Each section shows the panel big, then explains every column, icon and control in it.

The window at a glance



The whole window: menus and transport on top, the node canvas in the middle, the panel rail down the right edge.



The top bar: logo, File, Edit and Menu, play and rewind, tempo, time signature, key and scale, metronome, the audio switch and the mode badge.

CONTROL	WHAT IT DOES
Play / Rewind	Starts and pauses the shared clock that drives sequencers, LFOs and the arrangement; rewind returns to the start.

BPM (120.0)	Tempo. Double-click to type an exact value or drag to change it. Every tempo-synced node follows it.
Time signature (4 / 4)	Beats per bar and the note value of a beat. It sets the bar lines in the arrangement.
Key and scale (C Major)	The global key. Scale-aware nodes snap to it. The dropdown lists Major, Natural / Harmonic / Melodic Minor, the church modes, Major and Minor Pentatonic, Blues, Whole Tone and Chromatic.
Metronome	Click track on or off.
Start Audio / Stop Audio	Starts or stops the audio engine. While it is stopped you will hear nothing.
Timeline / Session / Studio badge	Shows which of them is driving playback right now.
fps, ms, cpu	Frame rate, frame time and audio load, at the far right when the window is wide enough. Hover the audio readout for dropout (xrun) counts.

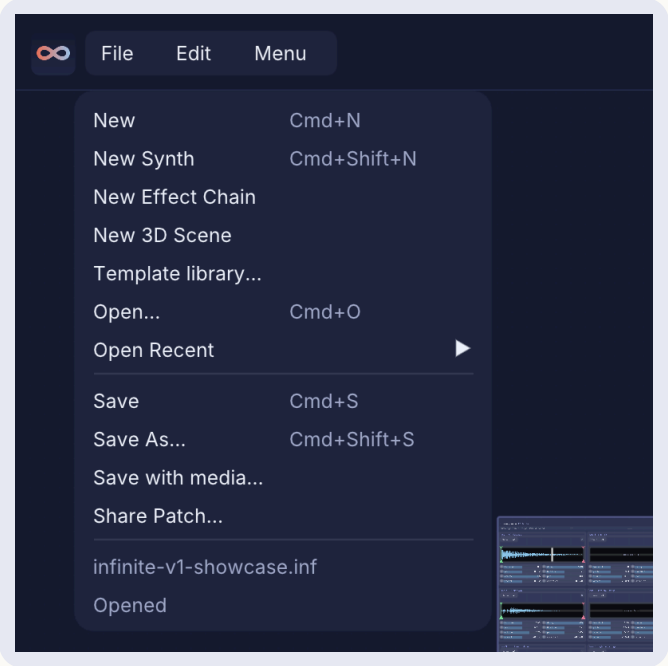
RAIL BUTTON	OPENS
Library	Every node, Field program, sample, media file and plugin.
Viewport	Live preview cards of any image node.
Mod Matrix	Every modulation route in one table.
Performance	Your own control surface of knobs, faders and pads.
Timeline	The arrangement: clips over time.
Session	Clip launcher: tracks across, scenes down.
Studio	Sample Studio: record, slice and export a take.
3D	3D View: the scene, its camera and gizmos.
Piano Roll	Note editing and Generate.
Terminal	A real shell inside the app, with the infinite command.
History (bottom)	Every edit, newest first.
Settings (bottom)	The Settings window.

Rail, top.

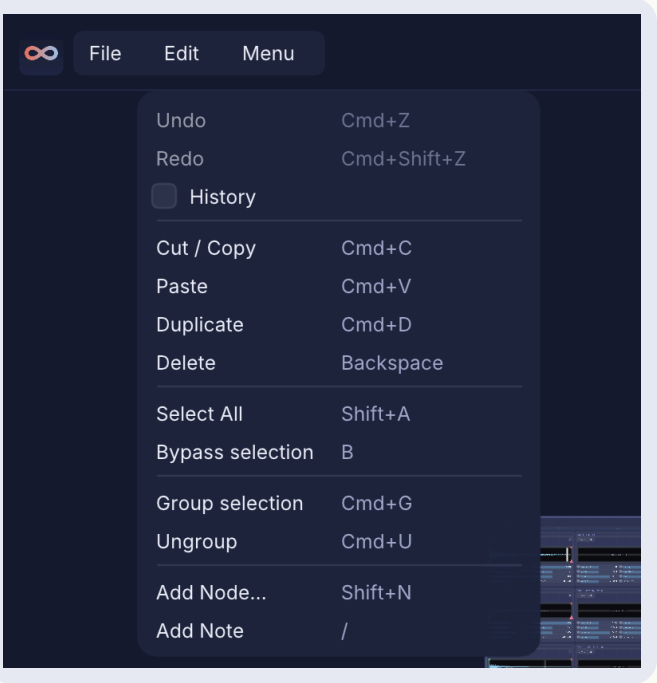
Bottom.

Menus

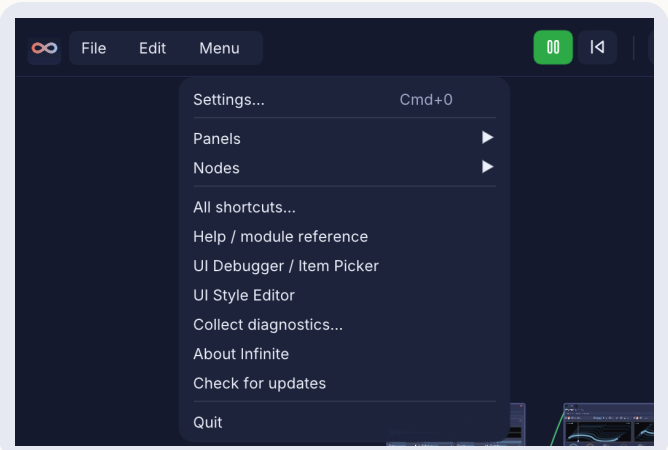
The three menus at the top left. File handles patches, Edit handles undo and clipboard, Menu holds the windows, help and the show/set tools.



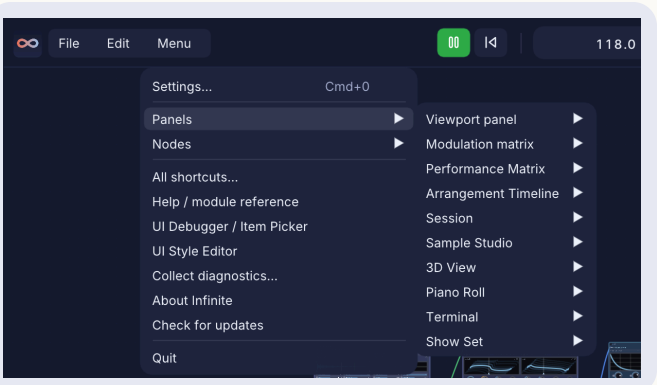
File



Edit



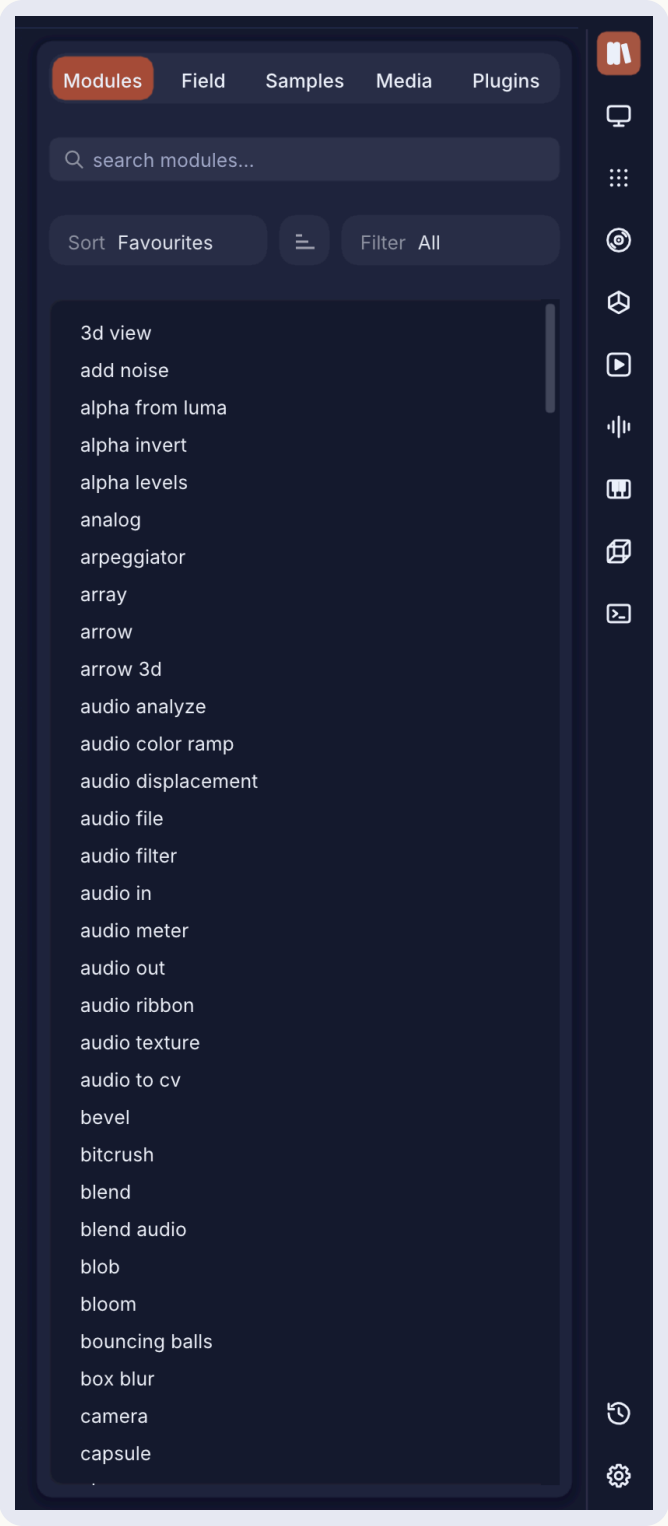
Menu



Menu, Panels: open any panel and set where it docks.

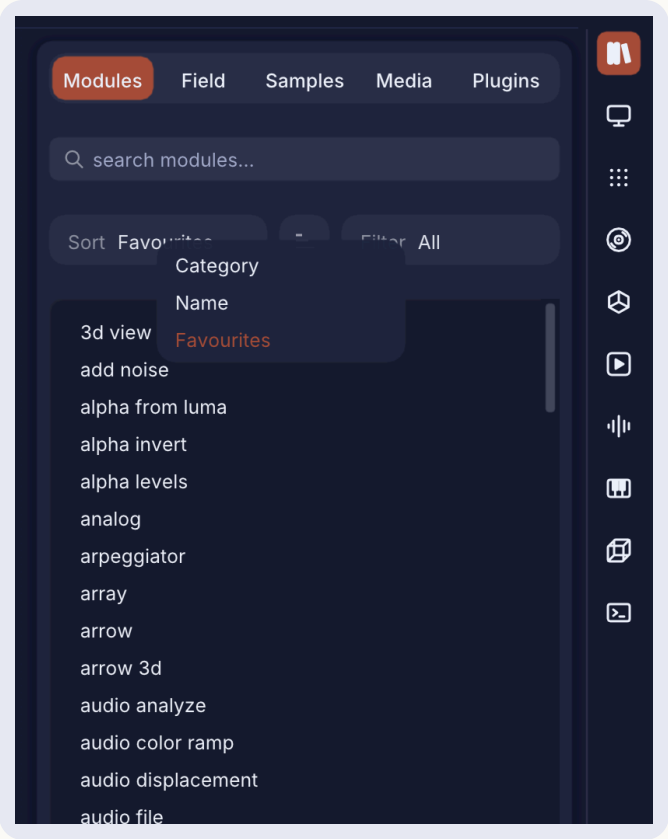
Library panel

The top button of the panel rail opens the library: every node, sample, media file and plugin you can drop on the canvas. Click an entry, or drag it onto the canvas.

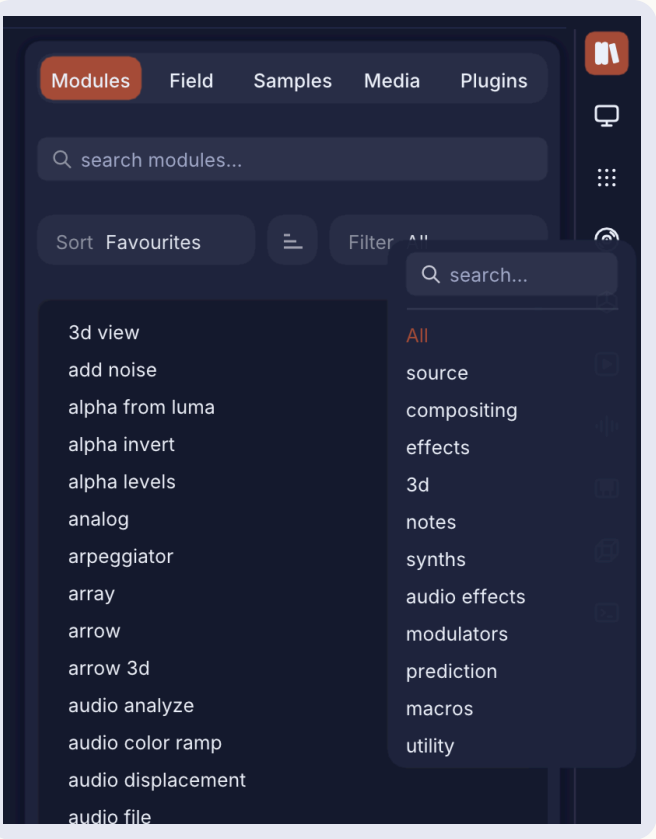


Library, Modules tab: search box, Sort, order toggle, Filter, then the alphabetical list.

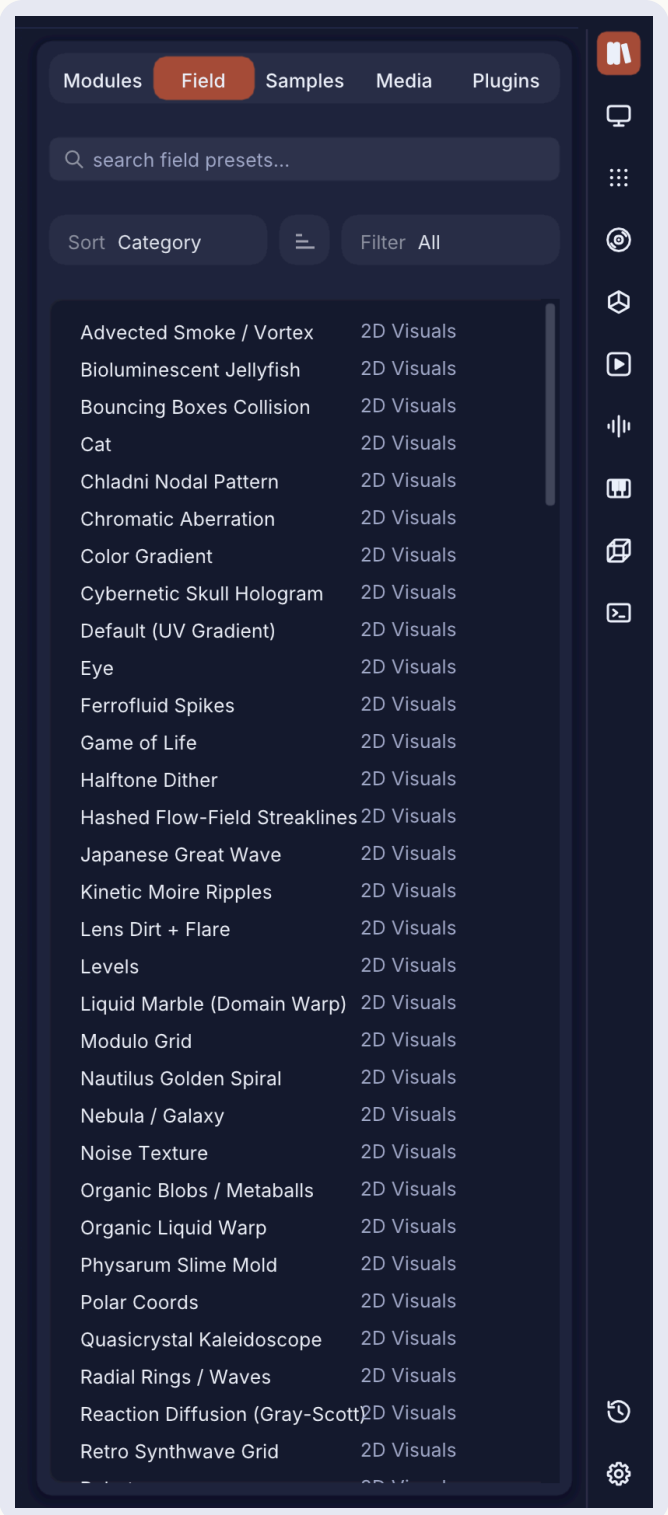
ELEMENT	WHAT IT DOES
Tabs	Modules (every node type), Field (Field-language nodes and examples), Samples and Media (your own files), Plugins (VST3 and AU instruments and effects).
Search	Filters the list as you type.
Sort	Orders the list: favourites first, alphabetical and so on. The button beside it flips ascending and descending.
Filter	Narrows the list by category, for example only effects or only synths.
List	Click to add at the centre of the view, or drag to a spot. Star an entry to make it a favourite.



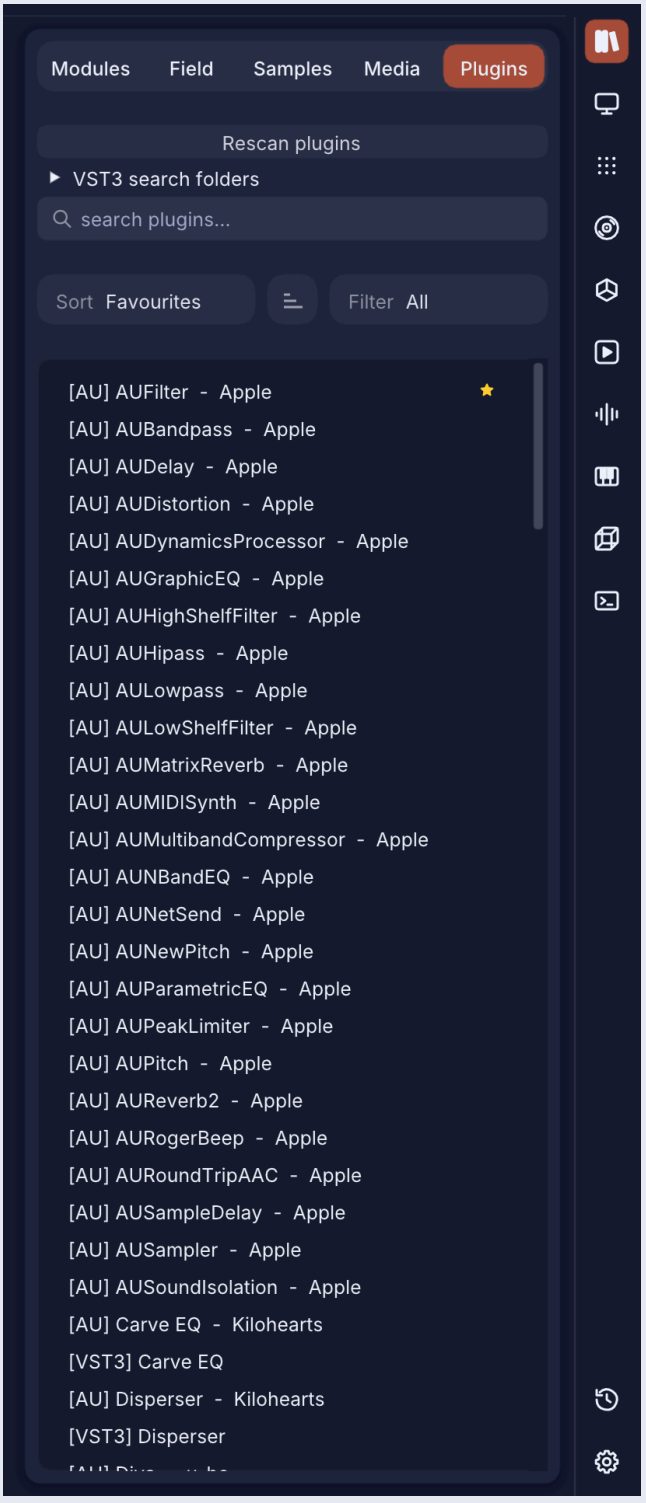
Sort dropdown



Filter dropdown



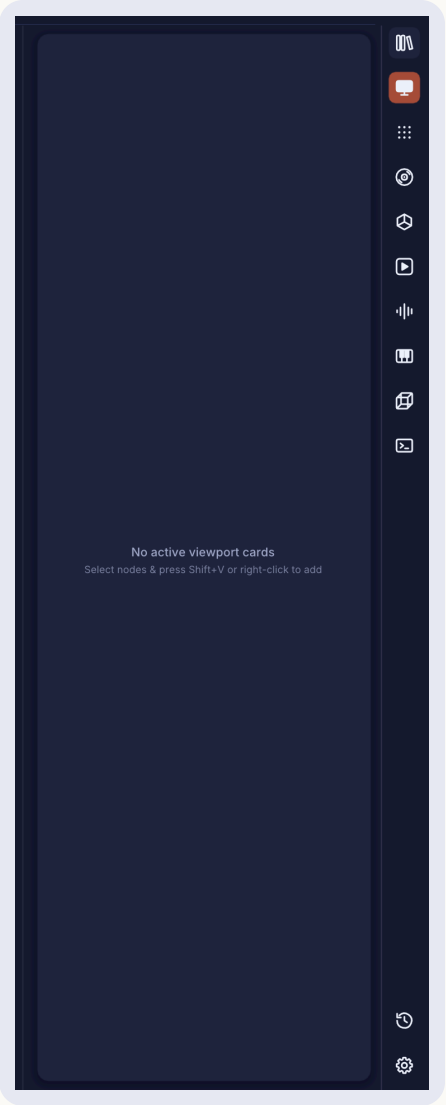
Field tab



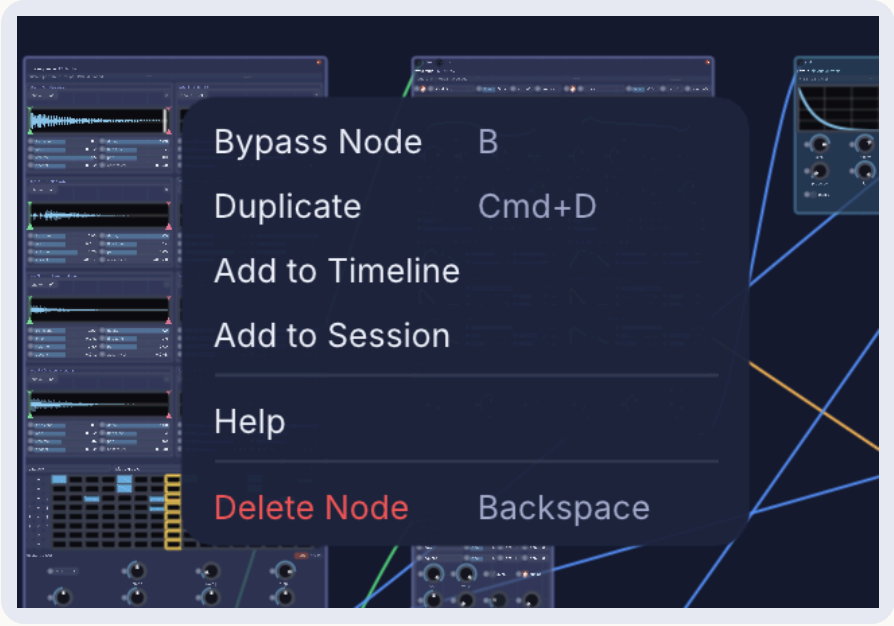
Plugins tab

Viewport panel

A live preview of any image node, next to the canvas. Right-click a node and choose **Open in viewport panel** (or select it and press Shift+V). Each card follows its node in real time and has a close button at its top right. Add several cards to watch different stages of a chain at once.



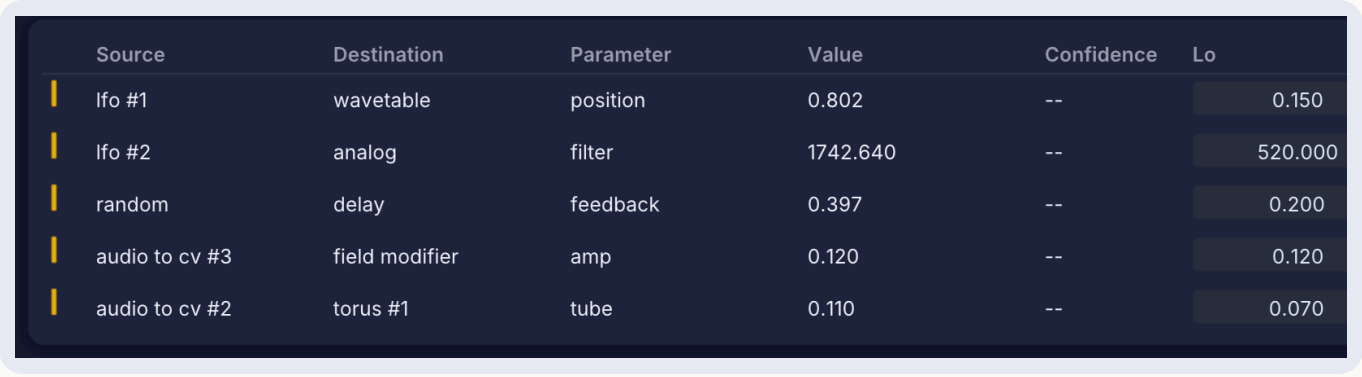
No cards yet: the panel says how to add one.



Right-click a node for its menu.

Modulation Matrix

Every modulation in the patch in one table: which source drives which parameter, and how. Drag the splitter above the panel to make it taller. The table is shown in two halves so each column is readable.



Left half: Source, Destination, Parameter, Value, Confidence.

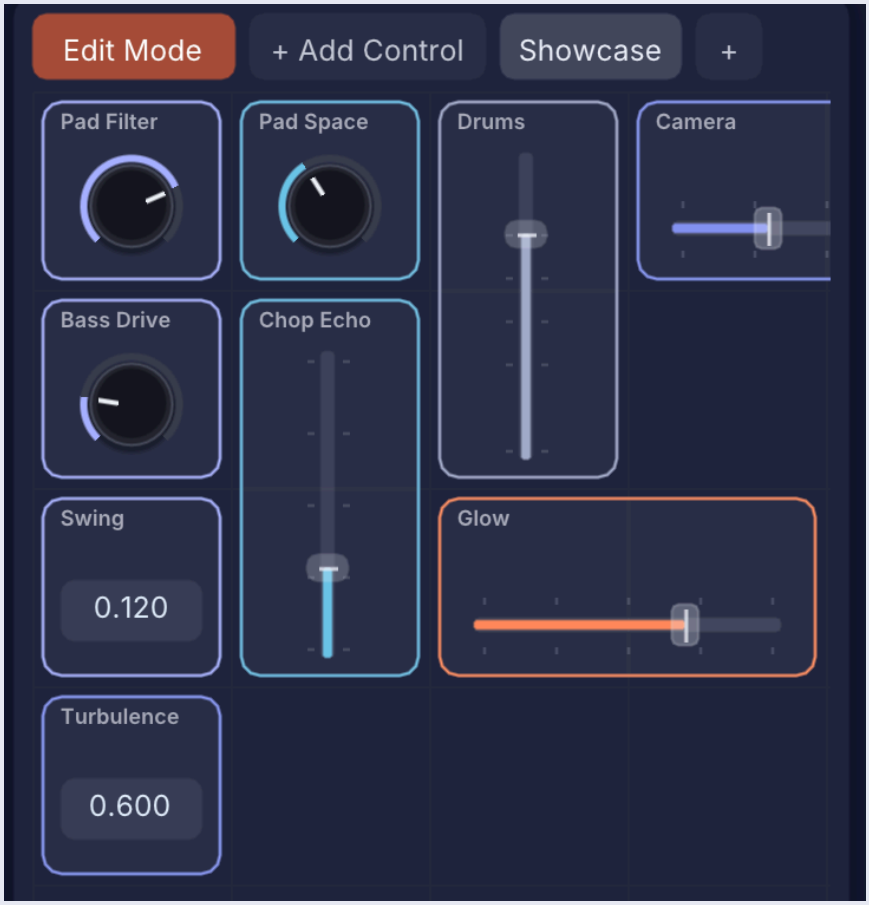
Right half: Lo, Hi, Inv, Curve, Signal and the delete button.

COLUMN	WHAT IT MEANS
Colour bar	The cable colour of the source, so you can match a row to the cable on the canvas.
Source	The node that produces the control signal: macro xy, random, an LFO, an envelope and so on.
Destination	The node that receives it.
Parameter	Which parameter on the destination is being moved.
Value	The live, current value of that parameter after modulation.
Confidence	Used by prediction sources to report how sure they are. It shows -- for ordinary sources.
Lo / Hi	The range the parameter travels between. Click a number to type an exact limit. Narrow it for subtle movement.
Inv	Inverts the mapping: when the source goes up, the parameter goes down.
Curve	The response curve between source and parameter, drawn as a small graph.
Signal	A tiny live scope of the incoming signal, so you can see it moving.
x	Removes the route. The cable on the canvas disappears with it.

Performance panel

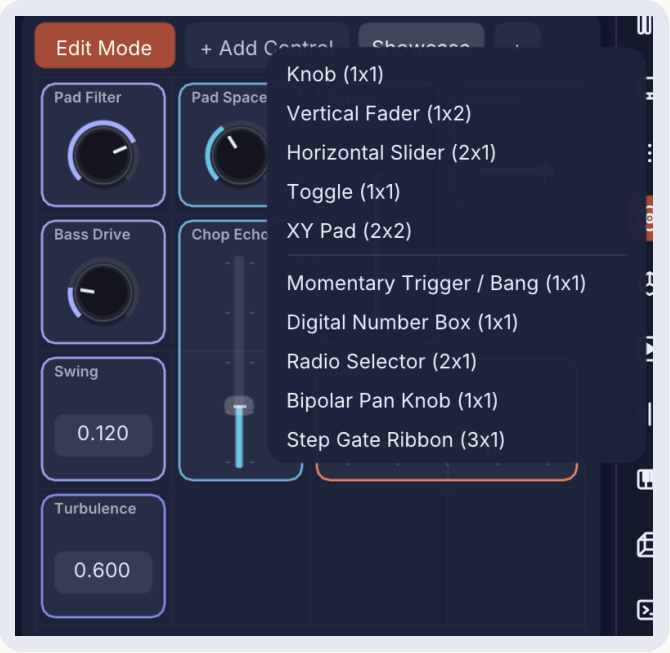
Build your own control surface from knobs, faders, pads and switches, then play it live. It opens in **Edit Mode**; switch off Edit Mode (the button then reads **Perform**) to lock the layout and play it. Drag the splitter up for more room.

The Performance panel docked at the right, in Edit Mode, with the showcase page.



Edit Mode: knobs, faders, number boxes and a slider on the Showcase page.

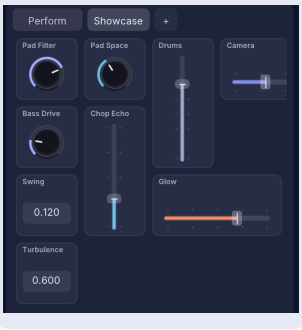
TOOLBAR	WHAT IT DOES
Edit Mode / Perform	Edit Mode lets you add, move, resize and rename controls. Perform locks the layout so a stray drag cannot move anything.
+ Add Control	Opens the control list below and places the chosen control on the next free cell.
Page tabs, +	Pages group controls into screens (here one page, Showcase). The + button adds a page.



+ Add Control menu: ten control types, with their grid size in brackets.

CONTROL	USE IT FOR
Knob (1x1)	Any single parameter.
Vertical Fader (1x2)	Levels and mixes.
Horizontal Slider (2x1)	Long throws, crossfades.
Toggle (1x1)	On/off switches and bypass.
XY Pad (2x2)	Two parameters with one finger.
Momentary Trigger / Bang (1x1)	Fires a one-shot while held.
Digital Number Box (1x1)	Type or drag an exact value.
Radio Selector (2x1)	Pick one of several steps, 1 to 8.
Bipolar Pan Knob (1x1)	Centre-detent knob for pan or detune.
Step Gate Ribbon (3x1)	Eight on/off steps for a gate pattern.

Right-click a control: Rename, Type, Assign Parameter, Clear Destination, MIDI CC Learn, Color Tint.



The same page in Perform mode: the toolbar shrinks to Perform and the page tabs.

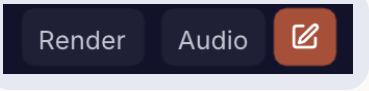
RIGHT-CLICK ITEM	WHAT IT DOES
Rename	Change the label shown on the control.
Type	Swap the control for another type without losing its assignment.
Assign Parameter...	Bind the control to a node parameter, then move the control to move the parameter. The current target is shown greyed below it.
Clear Destination	Unbinds the control.
MIDI CC Learn	Move a knob on your MIDI controller to bind it to this control.
Color Tint	Colour-code the control so groups are easy to find on stage.

Arrangement panel

The timeline. It sequences clips of nodes over time so a patch becomes a piece with a start and an end. Each icon in the toolbar is explained below; hover any of them in the app for its tooltip.

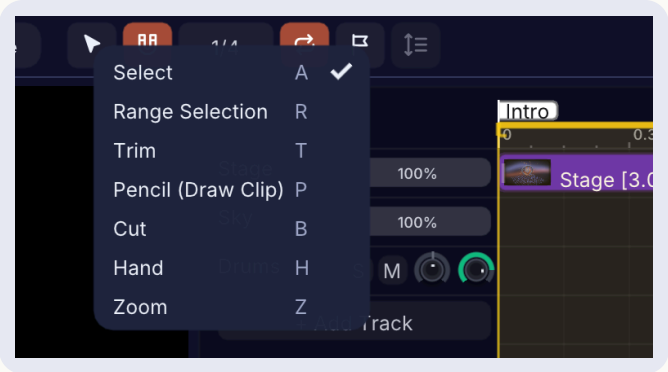
The Timeline docked at the bottom: monitor, tracks, clips and markers.

The Timeline made tall: toolbar, monitor, track list, markers (Intro, Groove), ruler, clips and the Inspector.

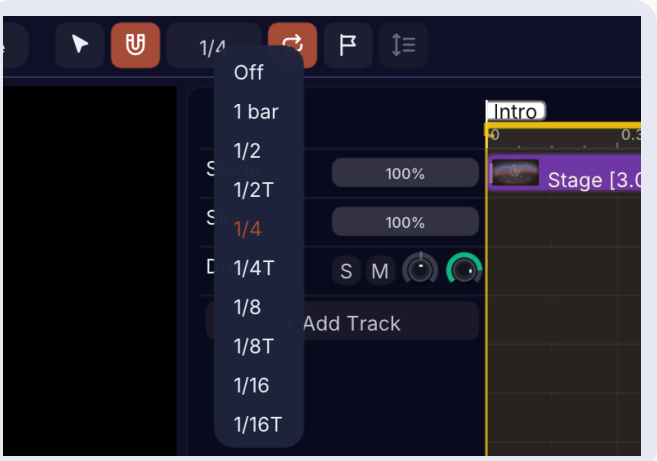


Right end of the toolbar: Render, Audio and the Inspector pencil.

TOOLBAR ITEM	WHAT IT DOES
Viewport monitor	Shows or hides the preview of the timeline output at the left of the panel. Right-click for its dock position.
Play / Pause (Space)	Plays the timeline from the playhead.
Return to Start (Enter)	Jumps the playhead to the beginning.
Bars / Time	Switches the ruler between bars and beats and minutes and seconds.
Tool (arrow)	Selects the editing tool. See the dropdown below.
Snap (magnet)	Clip edges snap to the grid when on.
1/4 (grid division)	How fine the snap grid is. See the dropdown below.
Loop	Turns the loop region on or off. Shift-drag on the ruler to set the region.
Flag (M)	Drops a marker at the playhead.
Row height	Resets any resized tracks back to the default height.
Render	Renders the timeline to a video file (or a WAV when it holds only audio).
Audio	Lets the timeline drive audio as well as picture.
Pencil	Opens or closes the Inspector.



Tool dropdown: Select (A), Range Selection (R), Trim (T), Pencil / Draw Clip (P), Cut (B), Hand (H), Zoom (Z).

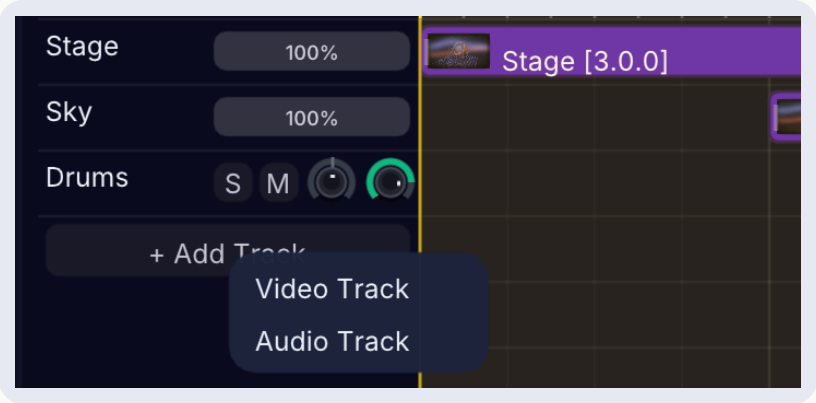


Snap grid dropdown: Off, 1 bar, 1/2, 1/4, 1/8, 1/16, each with a triplet (T).

Tracks, clips and the inspector

Adding tracks

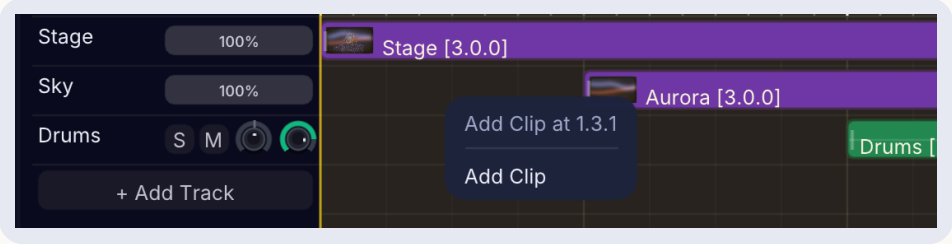
Click **+ Add Track** and choose Video Track or Audio Track. Audio tracks also carry **S** (solo), **M** (mute), a pan knob and a level knob; video tracks carry an opacity slider.



+ Add Track offers Video Track and Audio Track.

Adding a clip

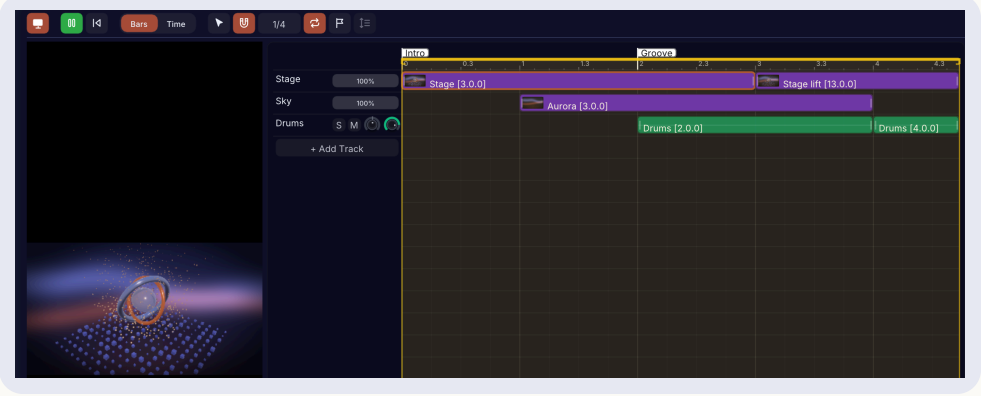
Right-click an empty spot on a track and choose **Add Clip**. The menu shows the bar and beat it will land on, snapped to the grid. A clip with no node yet is shown hatched.



Right-click a lane: Add Clip at the clicked bar and beat.

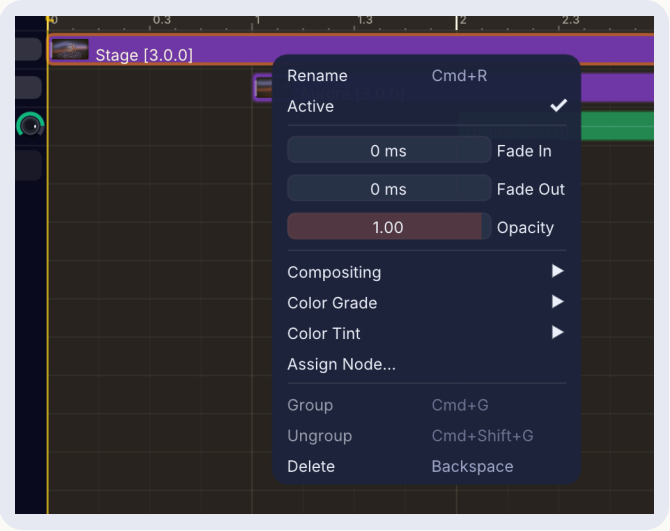
Assigning a clip to a node

Three ways: right-click a node on the canvas and choose **Add to Timeline** (creates an assigned clip at the playhead), choose **Assign Node…** in the clip's right-click menu, or use **Assign Different Node…** in the inspector. An assigned clip takes the node's name and a thumbnail, and the viewport monitor shows its output as the playhead crosses it.



The Stage clip selected (outlined); the monitor on the left shows the timeline output.

Right-clicking a clip

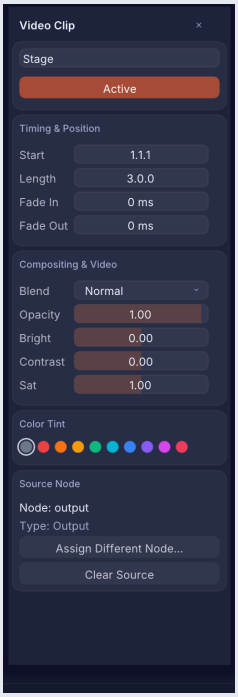


The clip menu.

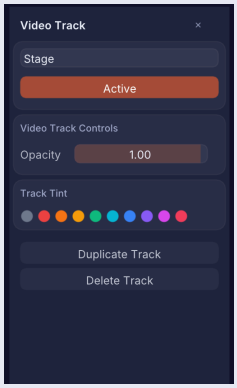
ITEM	WHAT IT DOES
Rename (Cmd+R)	Names the clip.
Active	Tick to play the clip; untick to mute it without deleting (it turns hatched).
Fade In / Fade Out	Fade length in milliseconds.
Opacity	Clip brightness against what is beneath it.
Compositing	Blend mode used to combine this clip with lower tracks.
Color Grade	A submenu of colour-grade adjustments for the clip.
Color Tint	Colour of the clip block on the timeline.
Assign Node...	Pick the node this clip plays.
Group (Cmd+G) / Ungroup (Cmd+Shift+G)	Treat several clips as one block you can move together.
Delete (Backspace)	Removes the clip.

The Inspector

The pencil icon opens a side panel for whatever is selected. Select a clip and it shows the clip; select a track and it shows the track.



Video Clip: name, Active, Timing & Position, Compositing & Video, Color Tint, Source Node.



Video Track: name, Active, opacity, Track Tint, Duplicate, Delete.

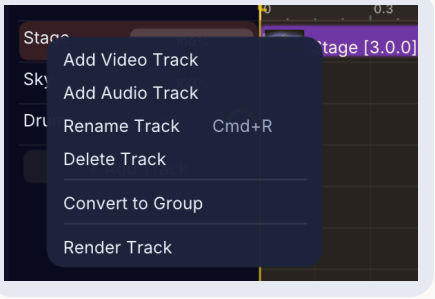
INSPECTOR FIELD	WHAT IT DOES
Name / Active	Rename and enable or disable the clip or track.
Start / Length	Exact position and duration in bars.beats.ticks. Type a value for sample-accurate placement.
Fade In / Fade Out	Edge fades in milliseconds.
Blend / Opacity / Bright / Contrast / Sat	How the clip composites over tracks beneath it.
Color Tint	The swatches colour the clip block.
Source Node	The node and node type the clip plays.
Assign Different Node… / Clear Source	Swap the node or make the clip unassigned again.

Selecting several clips and grouping

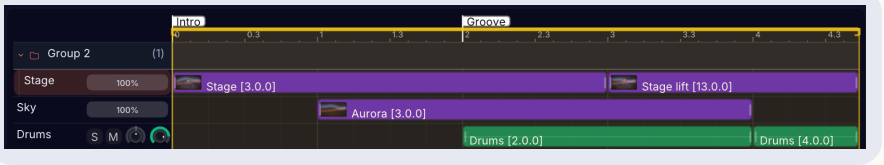
To move or colour several clips at once, press **R** for the Range Selection tool and drag a box over them, then press **A** to return to Select. Clips grouped with Group (Cmd+G) move as one.

Grouping tracks

Right-click a track name. **Convert to Group** wraps the track in a collapsible folder so a whole stack of tracks can be folded away. The same menu adds and renames tracks, deletes them, and offers **Render Track** to export just that track.



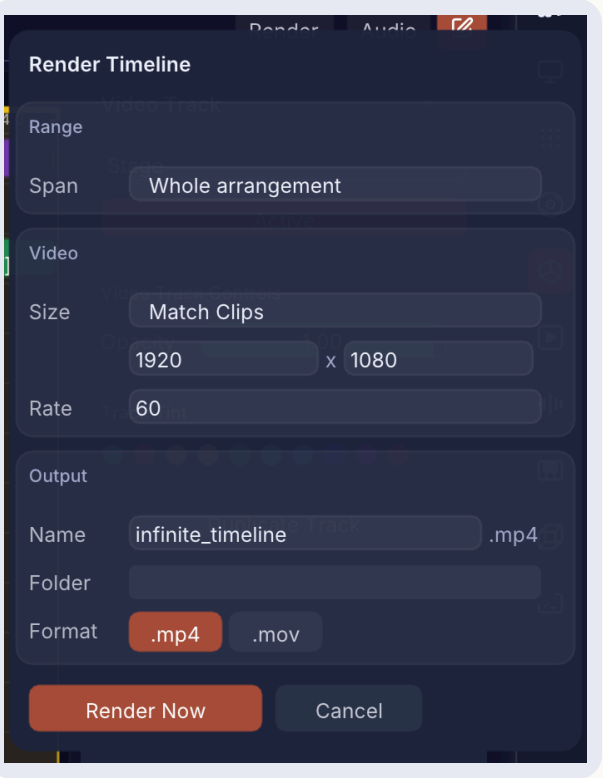
Track menu.



After Convert to Group: Group 2 (1) holds Stage.

Rendering

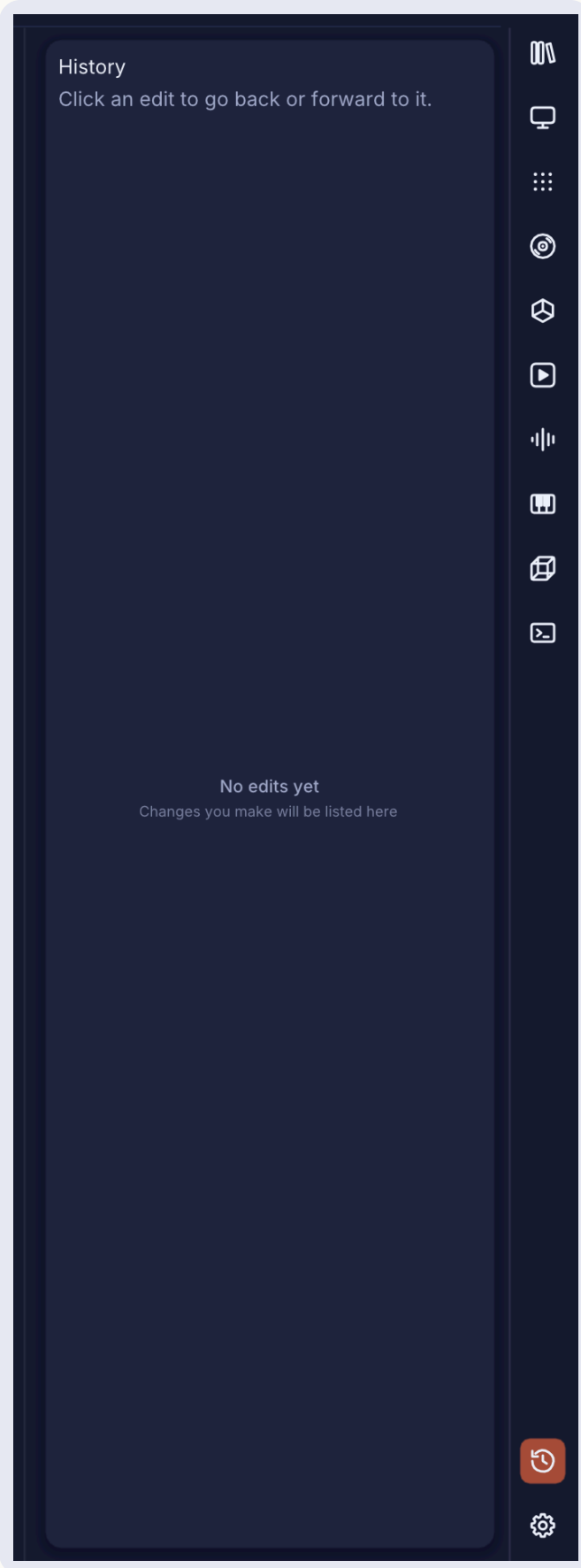
Render opens the Render Timeline popup: the span (whole arrangement, loop, markers or a custom range), the video size and rate, and the output name, folder and format.



Render Timeline popup.

History panel

Every edit you make is listed, newest at the top. Click any line to jump the patch back (or forward) to that moment; edits after it are kept until you make a new change.

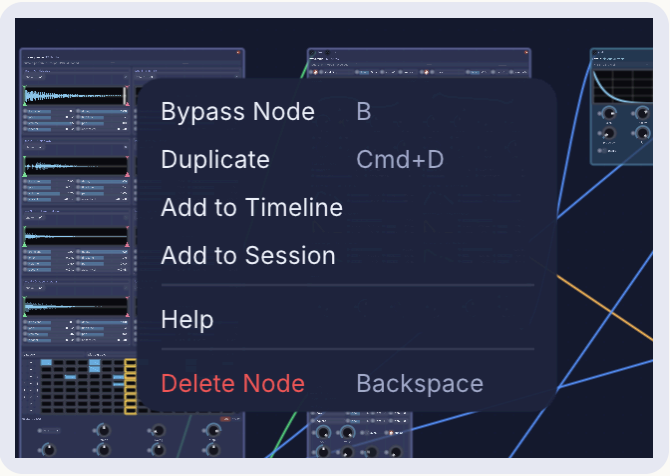


History before the first edit.

ENTRY	MEANING
Start	The patch as it was loaded.
Bypass	A node was bypassed or enabled.
Edit 2 settings	Two or more parameters changed together.
Edit timeline	A change in the arrangement.

Right-click menus and the Add Node palette

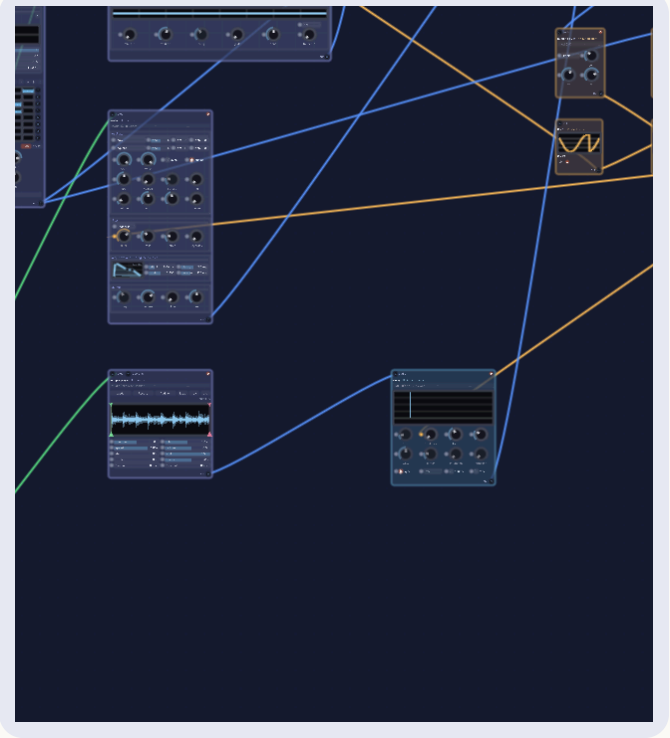
Right-click does different things depending on what is under the cursor.



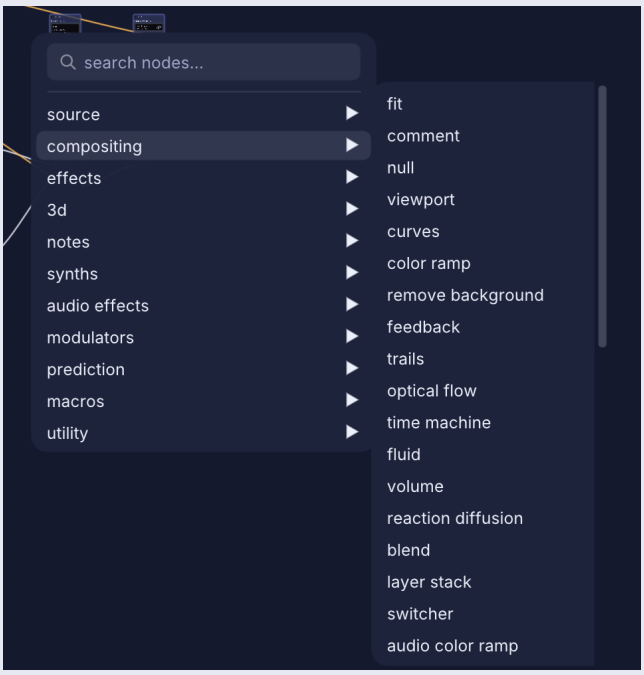
On an audio node: bypass, duplicate, add to Timeline or Session, help, delete. Image nodes add viewport and window items.

NODE MENU ITEM	WHAT IT DOES
Bypass Node (B)	Passes the input straight through.
Duplicate (Cmd+D)	Copies the node and its settings.
Hide params	Hides the node's parameter controls so it takes less room on the canvas.
Open in viewport panel	Adds a live card to the Viewport panel.
Add to Timeline	Creates an assigned clip in the Arrangement.
Add to Session	Puts the node in a Session slot.
Open in new window	Pops the node's output into its own window.
Help	Opens this node's help.
Delete Node (Backspace)	Removes it and its cables.

Right-click empty canvas, or press Shift+N, to open the **Add node** palette. Type to search, or walk the categories (source, compositing, effects, 3d, notes, synths, audio effects, modulators, prediction, macros, utility). The arrow means a submenu.



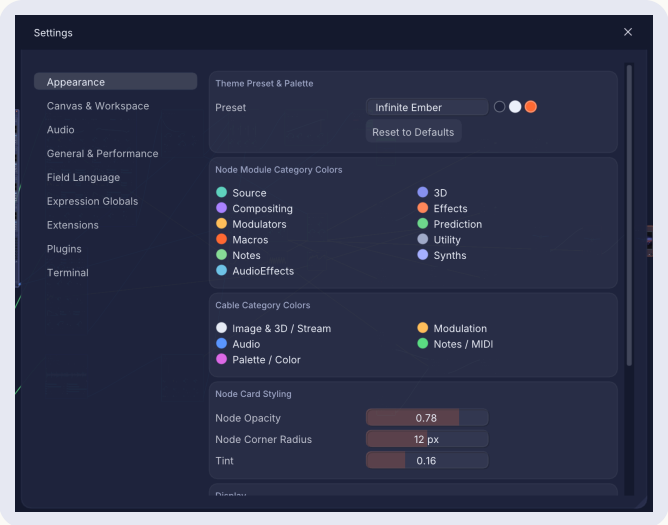
The Add node palette.



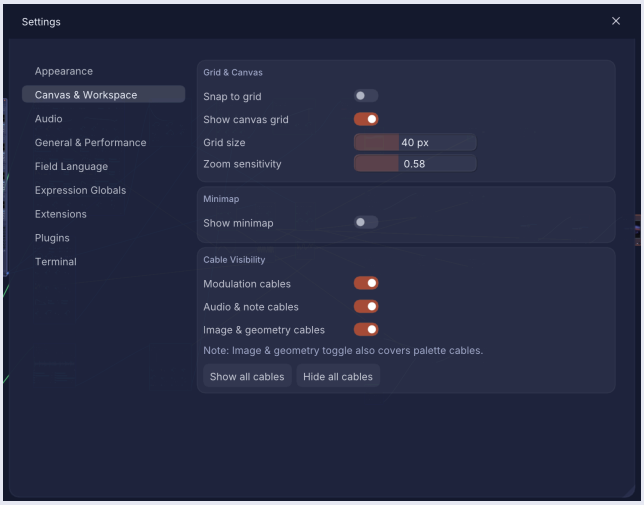
Hover a category to see its nodes: compositing shown.

Settings

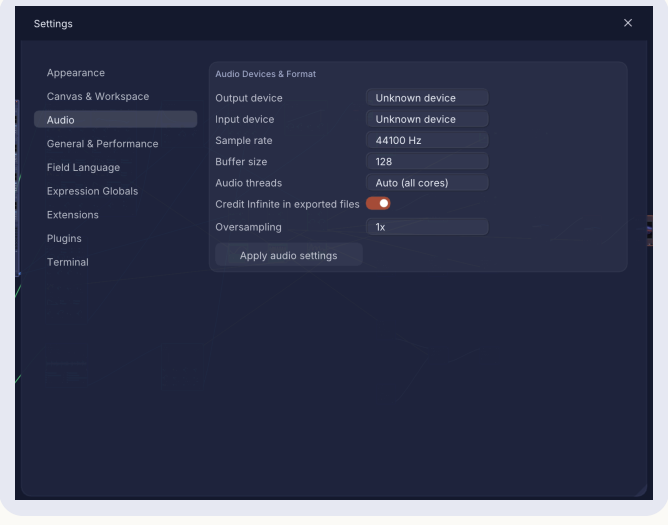
The gear at the bottom of the rail (or Cmd+0) opens Settings. It has nine pages.



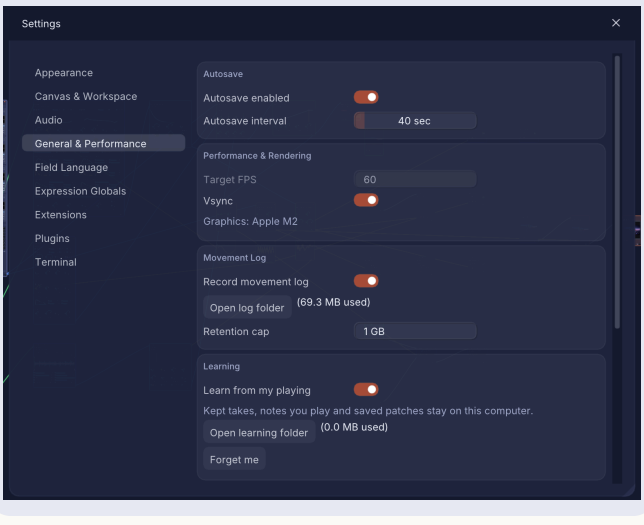
Appearance



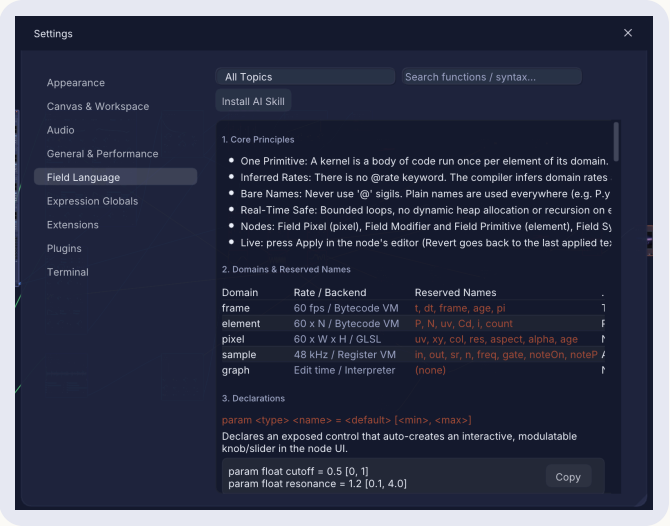
Canvas & Workspace



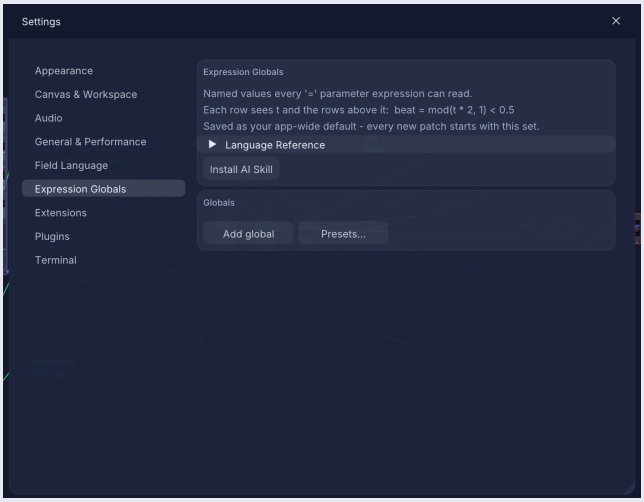
Audio



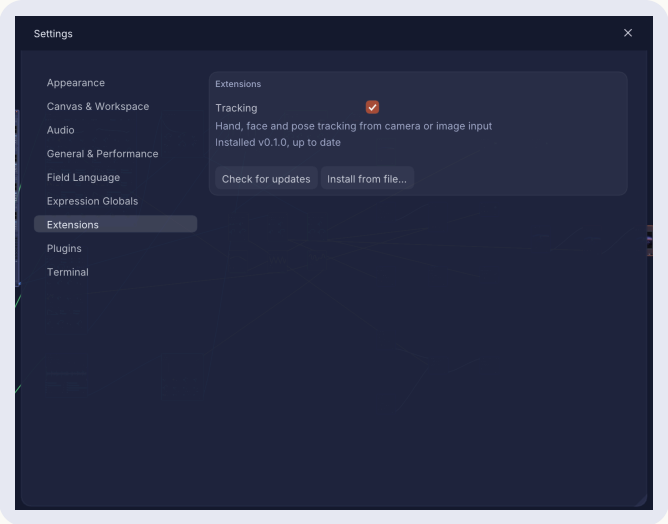
General & Performance



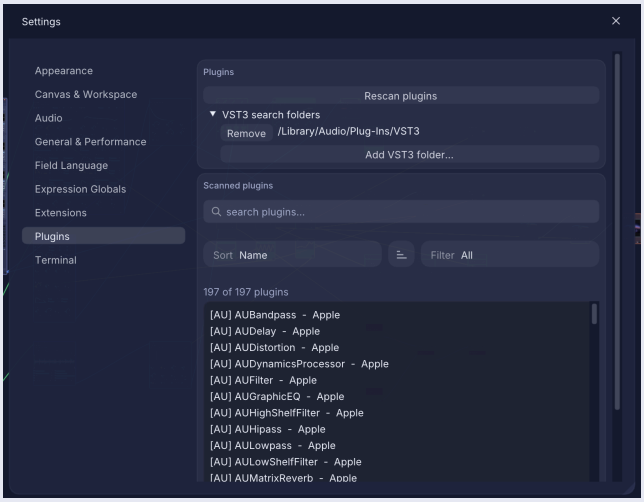
Field Language



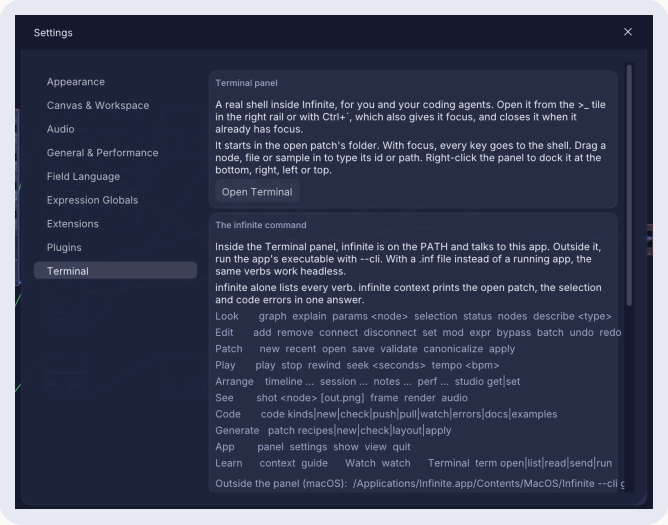
Expression Globals



Extensions: install extension packs.



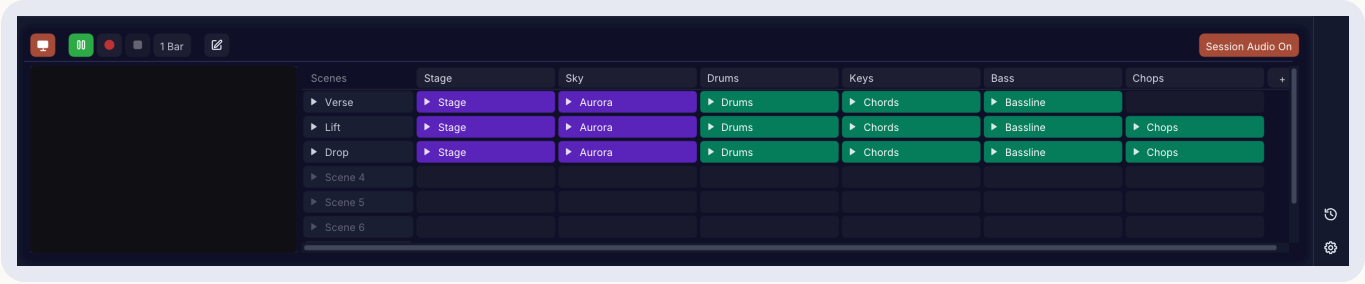
Plugins: VST3 folders and every scanned plugin.



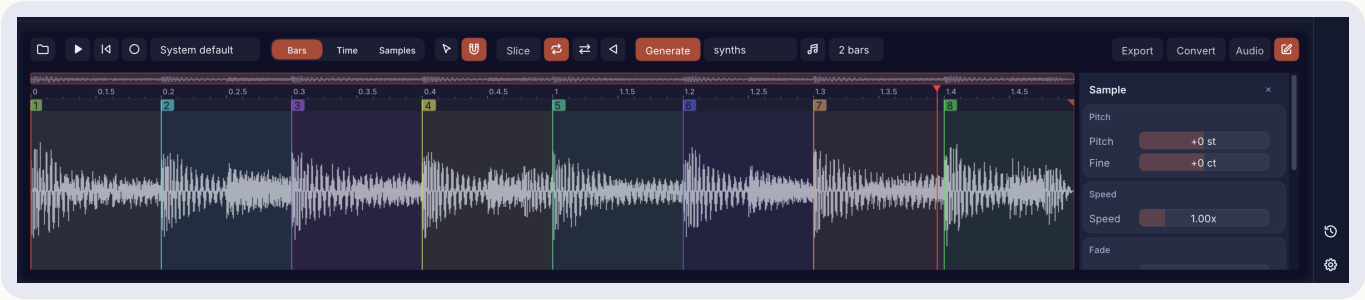
Terminal: the panel and the infinite command.

Sound panels: Session, Sample Studio, Piano Roll

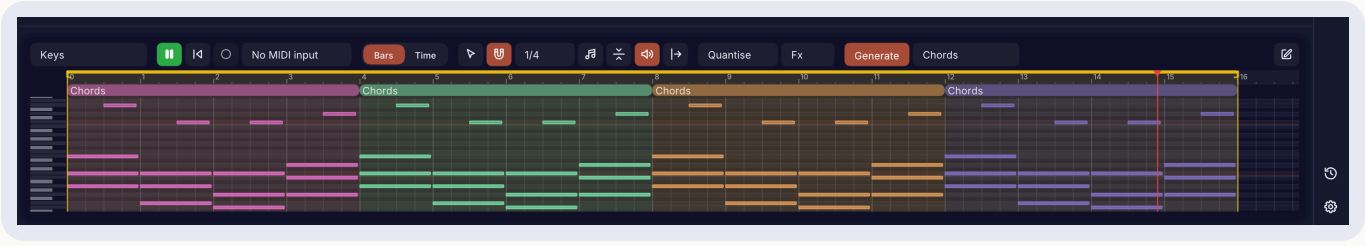
Three more strips dock along the bottom. Each has its own toolbar; right-click the toolbar for its dock position.



Session: tracks across, scenes down. Click a clip to launch it, click a scene name to launch the row. The playing scene is outlined.



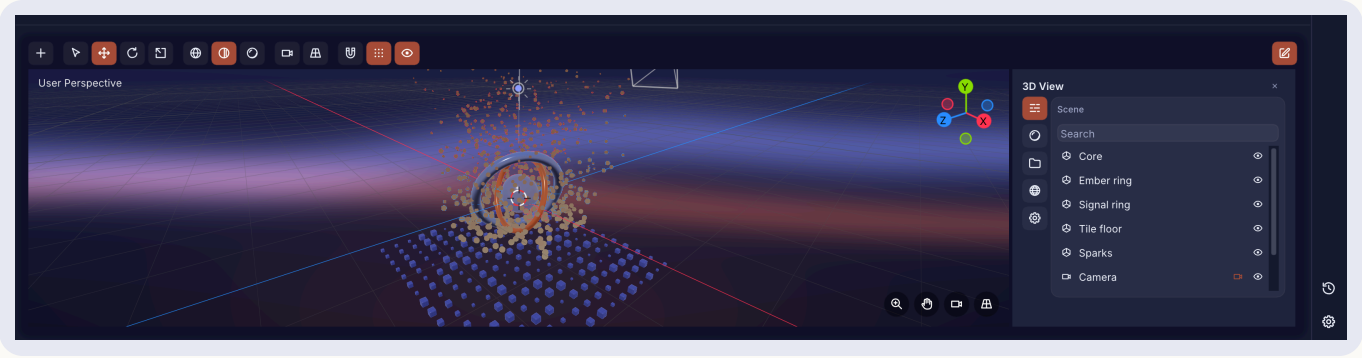
Sample Studio: a take cut into eight slices. Toolbar: open, play, input, Bars/Time/Samples, tools, Slice, Generate, Export, Convert. The Sample card on the right sets pitch, speed and fades.



Piano Roll: pages of notes over the keyboard, with record input, quantise, Fx and Generate in the toolbar.

3D View

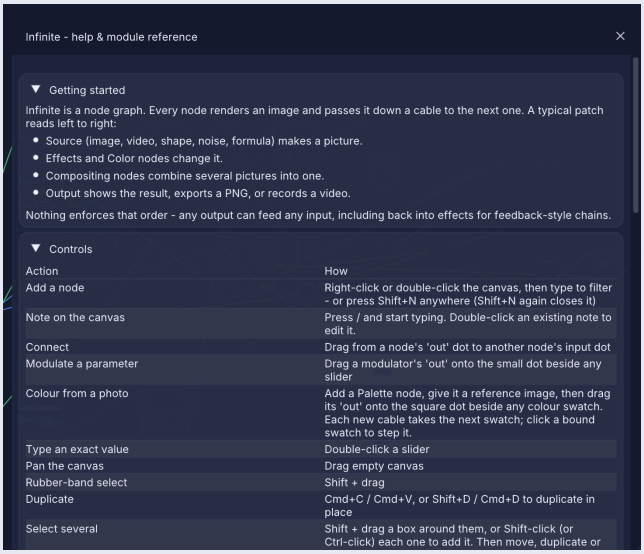
The 3D View shows the scene the 3D nodes build. The toolbar holds add, select, move, rotate and scale, the pivot and orientation choices and the view toggles. Drag the axis gizmo at the top right to orbit; the four buttons at the bottom right zoom, pan, fly the camera and lock it. The Scene list at the right lists every object; the eye hides one.



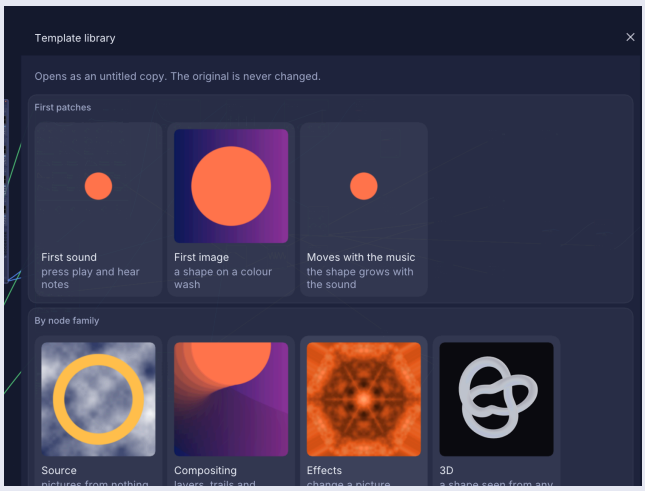
3D View docked at the bottom with the Scene list open.

Windows and dialogs

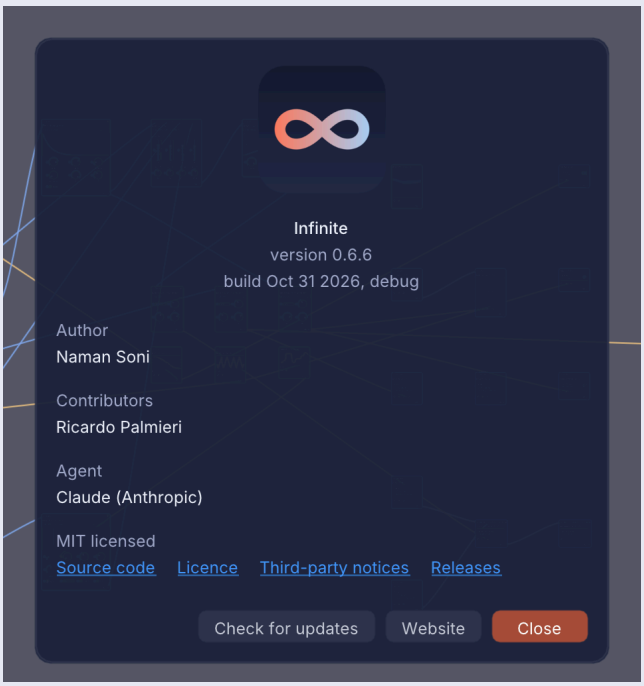
All Shortcuts (Menu, All shortcuts): filter box, then every shortcut by group.



Help and module reference: getting started, controls, then every node.



Template library (File, Template library): opens a copy; the original never changes.



About Infinite: version, build, credits, licence links.

Take your own screenshots

Every picture in this guide was made by script, so you can regenerate them after an update or make your own for a tutorial.

TOOL	WHAT IT DOES
scripts/screenshot_all_nodes.py	Renders every node type through the app's RPC `screenshot_node` and saves a 2x PNG for each into docs/node_screenshots/.
scripts/node_screenshot.py	Spawns one node by type name and saves a cropped screenshot of it. Useful for tutorials and bug reports.
tools/manual/ui_shots_offscreen.py	Runs the app off-screen and drives it over RPC (panels, synthetic clicks and keys) to capture the window, menus, panels, settings and dialogs in this chapter into docs/ui_screenshots/. Works with the screen locked.

```
python3 scripts/screenshot_all_nodes.py --out docs/node_screenshots
python3 scripts/node_screenshot.py "Wavetable" --out /tmp/shots
python3 tools/manual/build.py        # rebuild this guide with the fresh pictures
```

Screenshots are captured at 2x, show real values and are never recoloured. The control token for the RPC is read from the app's own settings folder, so nothing leaves your machine.

APPENDIX

Everything

Gallery: every node

One picture of every node type, grouped by palette category. Names are the exact type names the Infinite RPC and patch files use.

3D 93 NODES

Array

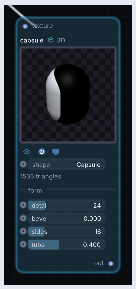
Arrow 3D

Audio Displacement

Audio Ribbon

Bevel

Camera



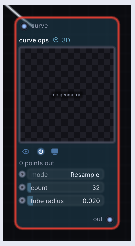
Capsule

Cloth

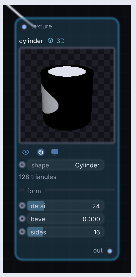
Cone

Cube

Curve



Curve Ops



Cylinder

Decimate

Delaunay Mesh

Delete

Delete Selected

Depth Projection

Difference

Disc

Displacement

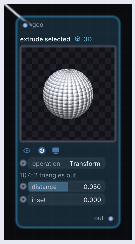
Distribute Points in Grid

Distribute Points on Faces

Dodecahedron

Explode

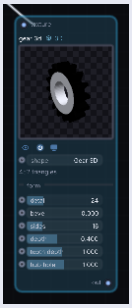
Extrude



Extrude Selected

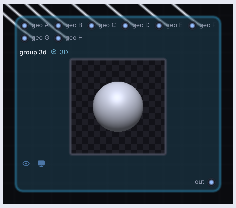
Field Modifier

Field Primitive



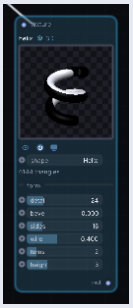
Gear 3D

Geometry



Group 3D

HDRI



Helix

Icosphere

Image to Points

Inset

Instance on Points

Intersect

Join Geometry

Klein Bottle

L-System

Light

Mapping

Material

Merge by Distance

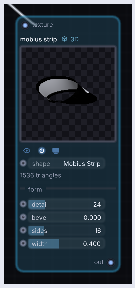
Mesh to Edges

Mesh to Faces

Mesh to Points

Metaballs

Mirror



Mobius Strip

Model 3D

Normals

Null 3D

Ocean

Octahedron

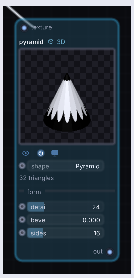
Particle System

Plane

Points to Vertices

Prism

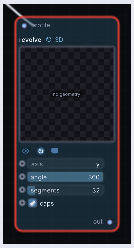
Proximity Lines



Pyramid

Render 3D

Resynthesize 3D



Revolve

Rigid Bodies

Rounded Cube

Screw

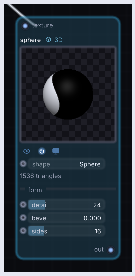
Select

Set Vertex Color

Sketch 3D

Smooth

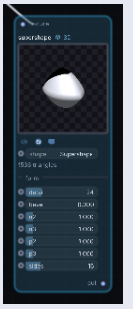
Solidify



Sphere

Star 3D

Subdivide



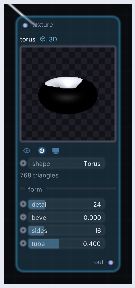
Supershape

Sweep

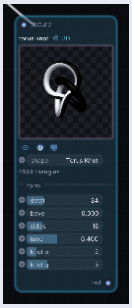
Switcher 3D

Tetrahedron

Text 3D



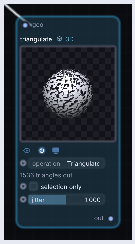
Torus



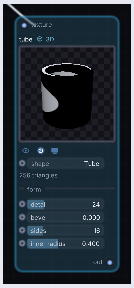
Torus Knot

Transform

Transform Selected



Triangulate



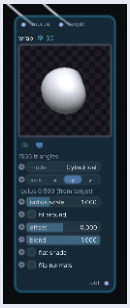
Tube

Twist

Union

Voronoi Cells

Wireframe



Wrap

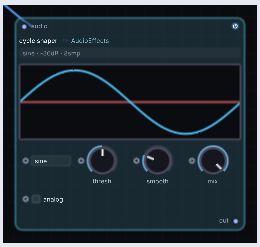
AudioEffects 30 NODES

Audio Filter

Bitcrush

Chorus

Convolution Reverb

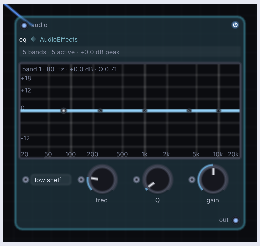


Cycle Shaper

Delay

Drive

Dynamics



EQ

Field Effect

Flanger

Formant Filter

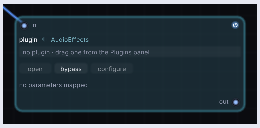
Frequency Shifter

Key-Snap

Limiter

Phaser

Pitch Shifter



Plugin

Resonator Bank

Reverb

Ring Mod

Saturator

Shape Resonator

Spec Blur

Spectrum Slide

Stereo

Stutter

Transient Shaper

Tremolo

Wavetable Shaper

Compositing 40 NODES

alpha from luma

alpha invert

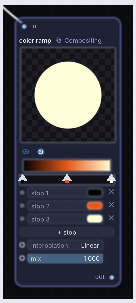
alpha levels

Audio Color Ramp

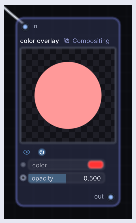
Blend

chroma key

color adjustments



Color Ramp



coloroverlay

Comment

Curves

dither

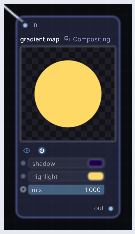
dropshadow

exposure

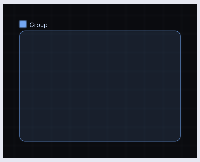
Feedback

Fit

Fluid



gradientmap



Group

invert

Layer Stack

lookup

luma key

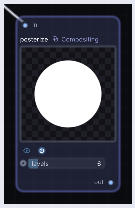
lut

Null

opacity

Optical Flow

outerglow



posterize

premultiply

Reaction Diffusion

Remove Background

set alpha

show alpha

Switcher

threshold

Time Machine

Trails

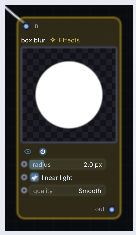
Viewport

Volume

Effects 28 NODES

addnoise

bloom



boxblur

convolve

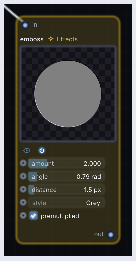
crop

diffuseglow

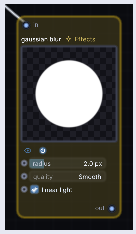
displace

edge outline

edge sobel



emboss

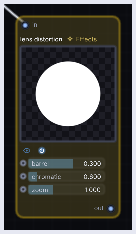


gaussianblur

glitch

halftone

kaleidoscope



lensdistortion

liquify

mirror tile

motionblur

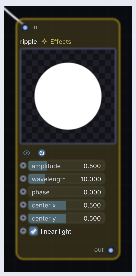
normal map

pinchpunch

pixelate

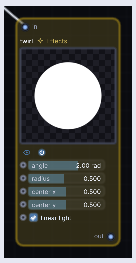
radialblur

Resynthesize



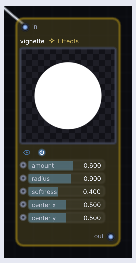
ripple

symmetry



twirl

unsharpmask



vignette

Macros 9 NODES

Macro Bipolar Knob

Macro Knob

Macro NumBox

Macro Radio Selector

Macro Slider

Macro Step Gate

Macro Toggle

Macro Trigger

Macro XY

Modulators 32 NODES

Audio Analyze

Audio File

Audio to CV

Compare

Constant

CV Recorder

CV to Pitch

Envelope

Face Track

Geometry Table

Hand Track

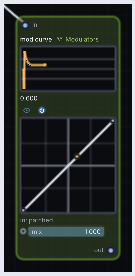
Image Analyze

LFO

Math

MIDI CC

MIDI Trigger



Mod Curve

Mod Depth

MSEG

Note Expression

Note to CV

Null Modulator

Palette

Path

Pattern

Pose Track

Random

Range to Range

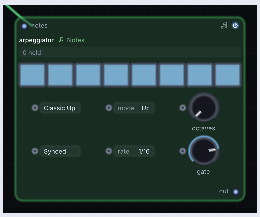
Smoothing

Velocity to CV

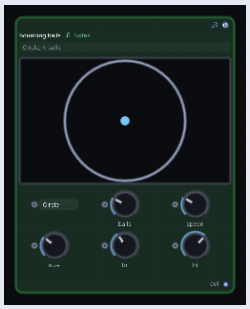
Vibrato

Voice Spread

Notes 24 NODES



Arpeggiator

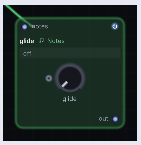


Bouncing Balls

Chorder

Field Notes

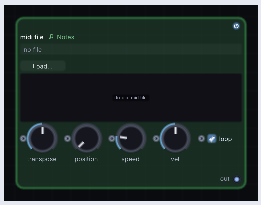
Gate



Glide

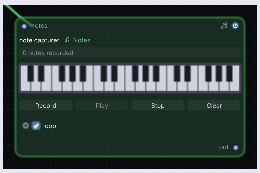
Humanizer

Keyboard



MIDI File

MIDI Notes



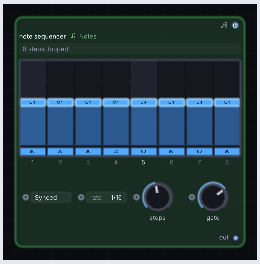
Note Capturer

Note Echo

Note Filter

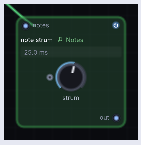
Note Merge

Note Router



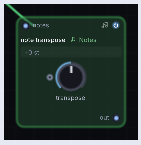
Note Sequencer

Note Stack



Note Strum

Note Switcher



Note Transpose

Pitch Bend

Quantizer

Random Note Generator

Velocity Curve

Prediction 8 NODES

Drift

Moves

Predictive Coloring

Predictive Modulator

Predictive Notes

Predictive Quantize

Predictive Rhythm

Predictive Velocity

Source 34 NODES

Arrow

Audio Texture

Blob

Chevron

Circle

Crescent

Cross

Draw

Ellipse

FieldPixel

Formula

Gear

Heart

Hexagon

Image Source

Line

Noise

Pie

Polygon

Ramp

Rectangle

Ring

Rounded Rect

Shape

Sketch

Slideshow

Star

Superellipse

Teardrop

Text

Texture

Triangle

Video

Video In

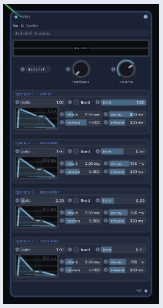
Synths 20 NODES

Analog

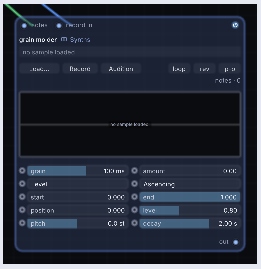
Drum Sequencer

Equation Synth

Field Synth



FM



Grain Molder

Granular

Looper

Metallic

Molder

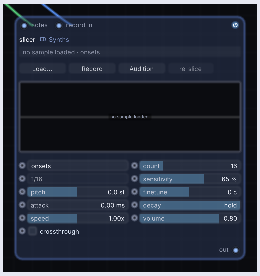
MPC

Oscillator

Partials

PaulStretch

Sampler



Slicer

Spectral Synth

String

Wave Terrain

Wavetable

Utility 18 NODES

Audio In

Audio Meter

Audio Out

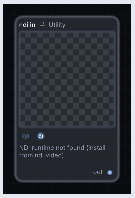
Blend Audio

Field Graph

Gain

MIDI Out

Mixer



NDI In

NDI Out

OSC Receive

OSC Send

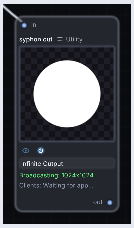
Output

Projection

Spatial Mixer

Splitter

Syphon In



Syphon Out

That's the guide.

333 node types, five cables, one canvas.

The best way to learn Infinite is to grab two or three nodes and see what happens when you wire them together.

For exact parameter ranges and engine internals, see the technical reference at n1m21n.github.io/Infinite. Need to script or automate Infinite? The Headless User Guide covers the command line and the control port, and the Field Language Manual covers writing your own nodes.